



INTEGRATED SCIENCE

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5
Class

FOCUS AREAS OF NEP 2020



Created with the new dimension of joyful learning based on **National Education Policy, 2020**.



Each chapter begins with **Put into Focus** includes concept



Key Words is a list of difficult words to help learners to understand the topics.



Cognitive Skills gives descriptive writing tasks to develop the written skills of the learners.



Critical Thinking is the ability to examine learner's thoughts and feelings to make decisions based on their life.



Observing Skills includes learning process whereby students learn with their own senses.





Communication Skills includes discussion which is generally a face-to-face interaction promotes confidence level.



Linking Chain
increases the ability
of students to excel
in two or more
different
subjects.



Activity includes varied activities in which learners construct and demonstrate understanding through an art form.



Computational Thinking provides learners problem solving technique.



Project work is the ability to do effective work together with others by doing some experiments.





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PREFACE

Science is an integral part of our lives. It is, therefore imperative that we help our children to learn the importance of science in such a way that it reflects through their attitude. This series '**Integrated Science**' aims to achieve the objective and to fulfil the guidelines of National Education Policy 2020. The series has been written in such a way as to kindle a genuine interest and motivation in a child to develop a scientific attitude and problem solving abilities from an early age.

This series focuses on the development of cognitive, effective and psychomotor skills through process based, activity-oriented learning, rather than through rote memorization of content-efforts have been made to help children in learning without burdening them, through simple text, child-friendly illustrations, pictures and ample activities. The aim is to encourage them to observe, interact, explore the world around them and learn by doing. Based on the experience gained through exploration, children are further guided and encouraged to **construct knowledge** in order to strengthen the foundation of their learning as well as develop language skills.

Wherever possible, the books of this series instil into children '**Values**' such as the importance of conserving nature, keeping health etc.

1

Living with Each Other

Piyush was merrily eating a rasgulla and watching television. A few drops of the rasgulla syrup fell on the floor. After a while, he noticed that an ant had appeared where the sweet syrup had fallen and then within minutes a large number of ants were moving in a line towards the syrup. He was amazed! How and from where did so many ants appear? They must have a fantastic way of communicating. His father, who was watching Piyush's observation, smiled and told him that ants are social insects who live and work in groups. When they find food, they leave behind **scent trails**, to let other ants know about the food source.



Put into Focus

- ▶ Social insects like ants, wasps, termites and bees live in colonies.
- ▶ They share work and resources.



Inside an ant hill

Ants

Ants often seen carrying food and other material to their nests, which may be underground or on top of the ground.

An ant's nest is like a small **colony** where hundreds of ants live together. Ants usually dig underground and their nest consists of tunnels, store rooms and chambers, which are used for laying eggs or storing food. Some ants also build mounds above the ground and cover them with twigs and leaves.

Most ant colonies consist of a **queen ant**, **worker ants**, **soldier ants** and **drones**.



Drones

Drones are male ants. The queen is the largest ant in the colony. Her job is only to lay thousands of eggs.

The queen ant lays eggs with the help of drones. The **worker** ants take care of the eggs. Eggs hatch into **larvae** that grow



Queen ant

into ants. Worker ants get food to feed the larvae as well as the queen ant. Workers also bring leaves and twigs to build and repair. The **soldier ants** are bigger than worker ants. Their job is to protect the nest.

Honeybee's nest

Just like ants, honeybees too live in groups and work together. Wild honeybees make nests on trees, while in farms they are reared to obtain honey. They live in a beekeepers hive. The honeybee's nest or hive is a collection of chambers that contains many hexagonal (six-sided) cells. These are made up of wax that comes from the bee's bodies. These cells are used for laying eggs, raising the young ones and storing food.



Know More

Ants are intelligent. Scientist have discovered that the ant's brain can function as fast as a computer.

Raising the young

Honeybees too have a well-organised disciplined community where different groups of bees do different jobs.

Queen bees is the head of a colony. She is bigger than the other bees and her only job is to lay eggs. She moves around the hive and lays an egg in each empty cell. A queen bee is called so because she does no other work. **Worker bees** are responsible for the cells, finding foods and feeding the larvae.



Inside a beehive

Male bees are known as **drones**. They mate with the queen bee for producing eggs.

Worker bees have a sharp sting. They possess pollen baskets, wax secreting glands and a honey sac for collecting honey.



Making Honey

Making honey

As a worker bee sits on a flower, it takes out its tube-like tongue that sucks up the sweet nectar. At the same time, some pollen too falls in the pollen baskets of the honeybee. When it sits on another flower, some pollen rubs off and is used to make seeds. This is known as **pollination**.

It is very important natural process in which the honeybees prove extremely helpful. Worker bees also take some pollen for themselves and mix it with nectar to make honey. Honey is stored in cells that do not contain larvae.

Interesting facts about honeybees

Honeybees have a very interesting way of communicating with other bees. When they find food and need to inform the other bees, they dance in a particular manner. If the

food is nearby, they perform a round dance and if it is far, they do a waggle dance. The extent of wagging tells the other bees how far is the source of food.



Know More!

Ant can lift weight which is 20-50 times their own body weight.



KEY WORDS

Pollen: Pollen is a fine powder produced by flowers.

Nectar: Nectar is a sweet liquid produced by flowers, which bees and other insects collect.

Colony: A community of animals or plants of one kind.



OVERVIEW ON THE CHAPTER

▶ Ants and bees are examples of social insects.

- ▶ Social insects live together in groups. They share work and resources.
- ▶ Ants live in a nest or colony that contains many rooms, chambers and tunnels.
- ▶ An ant colony consists of a queen ant, worker ants, drones and soldier ants.
- ▶ A queen ant lays eggs while worker ants get food and build nests. Soldier ants guard the colony. Drones help the queen to produce eggs.
- ▶ Honeybees live in a nest or a hive. The beehive is made up of hexagonal cells that store food and keep eggs.
- ▶ A queen bee lays eggs, while the worker bees take care of the eggs, and get food for the larvae as well as the queen bee.



Assignment



COGNITIVE SKILLS



Multiple choice questions :

(Learning by Doing)

Tick (✓) the correct option.

1. _____ is the head of a colony.

a. queen bee

b. worker bee

c. drone

d. soldier ant

2. Male bees are known as _____.

a. queen bees

b. drones

c. worker bees

d. worker ants

3. _____ and _____ are examples of social insects.
- a. ants ☐ b. bees
- c. butterfly ☐ d. none of them

 Give one word for the following :

1. Largest ant in the colony that lays eggs.
2. Nest of a honeybee.
3. Sweet juice from the flowers sucked by worker bees.
4. Transfer of pollen from one flower to another.
5. Young ones of ants and bees that hatch from their eggs.

C Match the column A with the column B :

- | A | B |
|----------------------------|-------------------------------------|
| 1. Yellow powder substance | a. sweet substance made from nectar |
| 2. Soldier ant | b. clean the cells |
| 3. Honey | c. shape of the beehive cells |
| 4. Hexagonal | d. pollen |
| 5. Worker bees | e. guard the nest |

Answer the following questions :

1. Why are ants and bees called social insects?

2. How is the ant's nest different from the hive of a honeybee?

3. What is the similarity between an queen ant and a queen bee?

4. Explain how honeybees help in pollination.

CRITICAL THINKING

(Self-Awareness)

1. Fill in the blanks according to the kind of work done by various kinds of ants.
- _____ ants protect the nest.
 - _____ ant lays eggs.
 - _____ ants collect food.

4. _____ ants repair and build the nest.
5. _____ fertilisers the eggs laid by the queen ant.

OBSERVING SKILLS

Label the parts of a Honey bee.

(Perceptual Thinking)



COMMUNICATION SKILLS

(Discussion)

Teacher may arrange a visit to a bee farm. See how honey bees are reared to produce honey. Speak to the beekeeper. Ask him the following questions and record here or make a project file.

- | | | |
|---------------------------------------|---|-------|
| 1. Beekeeper's name | — | _____ |
| 2. Name of the farm | — | _____ |
| 3. Number of beehives | — | _____ |
| 4. Amount of honey produced in a week | — | _____ |
| 5. Rate at which he sells the honey | — | _____ |

2

Balance in Nature



Green Forest

In a lush green forest, a deer is happily moving around and eating the soft green plants.

Suddenly, a leopard which had been waiting patiently in the bushes leaps out and catches the deer-its food!

Life in any habitat begins with the sun. Without the sun, plants cannot make food.

Sun energy is used by plants to convert water and carbon dioxide into starch that is stored as food. This process is called **photosynthesis**. It also helps the plants to grow. Since plants are the only living things that can make their own food, they are called nature's **primary producers**.

The primary and secondary consumers

Green plants are eaten by many kinds of animals. The rabbit, deer, squirrel as well as all the grazing animals such as cow, sheep and goat are plant eaters. They are known as **herbivores**.

All these animals, which eat food made by plants, are called **primary consumers**.



Rabbit



Deer



Squirrel



Put into Focus

- ▶ Food chain in a habitat
- ▶ Producers, primary consumers and secondary consumers
- ▶ Food web
- ▶ Keeping the balance in the nature



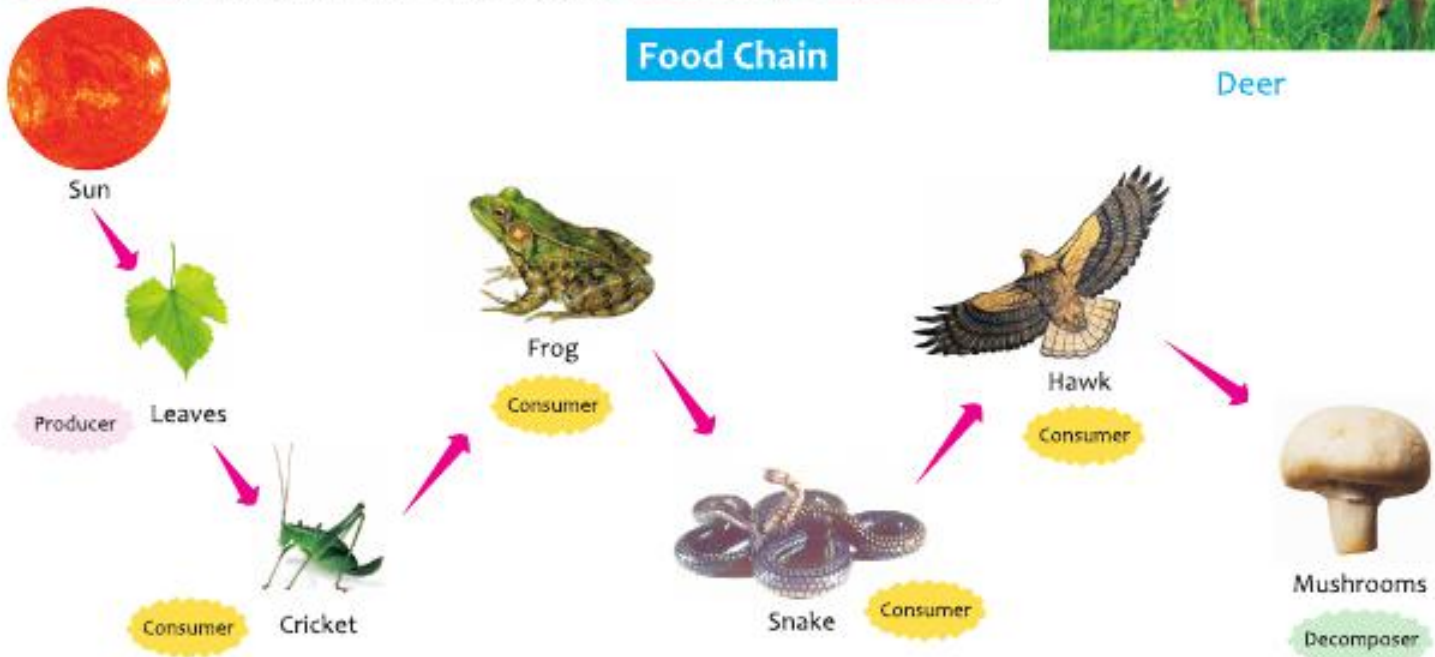
Sun

The primary consumers such as rabbit, caterpillar or deer become food for animals that hunt other animals like shark in the ocean or a lion in the forest.

Animals that feed on the flesh of other animals are called **carnivores**. Carnivores are also called **secondary consumers**.



Deer



The plants are eaten by herbivores and herbivores are eaten by carnivores. This is not the end of the chain. When the carnivores die they are eaten by **scavengers**.

The relationship between the plants and the animals that eat them is called a **food chain**. It follows a single path, as animals find food.

Grain, a primary producer, is eaten by a mouse which is a primary consumer. Owl that eats the mouse is the secondary consumer.

Food web

When you eat potato chips you are part of one food chain, when you eat fish it is another food chain. Though all food chains may appear to be separate, they are actually interconnected in a large and complex **food web**. Food webs show how plants and animals are interconnected in many ways to help them all survive.

Keeping the balance in the nature

Food chains are a way of keeping the balance in nature. Each link in the food chain is important. Look at the simple food chain.

If lion is hunted excessively by man and their number goes down what will happen?



The deer population will increase tremendously. When deer will be born in large numbers and live freely, they will eat away all plants and grass and then the forest will become a desert.

Dependence of plants and animals

Each animal and bird species is important to maintain nature's balance. Animals and plants depend on each other. Without plants there would be no food and oxygen. Without animals, plants would not get carbon dioxide for photosynthesis. Many insects play an important role in pollination of seeds. Carnivores help to prevent many herbivores population from increasing excessively. Hawks and eagles clean up environment by eating dead animals.

Both plants and animals depend on man to let them live. By cutting down the forests, we destroy the plants as well as homes of various animals living in them. When humans hunt some animals excessively, their population decreases and after some time they disappear completely.



Know More!

The Sahara desert is expanding southwards, engulfing grasslands at a rate of 30 miles (50 km) every year.



KEY WORDS

Herbivores: An animal that feeds on plants.

Carnivores: An animal that eats meat.

Photosynthesis: The process by which green plants use sunlight to form nutrients from carbon dioxide and water.



OVERVIEW ON THE CHAPTER

- ▶ In a particular habitat, many plants and animals live together.
- ▶ They are interlinked with each other by a food chain.
- ▶ The relationship between the plants and animals who eat them is called a food chain.
- ▶ The food chain begins with plants as primary producers of food.
- ▶ Plant-eating animals or herbivores are the primary consumers.
- ▶ Flesh-eating animals or carnivores form the second step of the food chain. They are secondary consumers.
- ▶ Many food chains are interconnected. They form a food web.
- ▶ Each link of the food chain is important to keep the balance in nature.
- ▶ Plants and animals are dependent on each other.
- ▶ Some of man's actions disturb the balance of nature.

Assignment



COGNITIVE SKILLS

(Learning by Doing)

A Multiple choice questions :

Tick (✓) the correct option.

- _____ eating animals are the primary consumers.
a. plant ☐ b. flesh ☐ c. bird ☐ d. insect ☐
- _____ eating animals are the secondary consumers.
a. plant ☐ b. flesh ☐ c. bird ☐ d. insect ☐
- _____ and _____ are dependent on each other.
a. plants ☐ b. animals ☐
c. both (a) and (b) ☐ d. none of them ☐

B Write 'T' for true and 'F' for false :

- Plants are called secondary consumers.
- The interconnection between many food chains forms a food web.
- Food chains disturb the balance of nature.
- Flesh-eating animals are the second step of food chain.
- Grain → mouse → owl. This is known as food web.

☐
☐
☐
☐
☐

C Match the column A with the column B :

A

- Primary producer
- Carnivores
- Herbivores
- Food chain
- Food web

B

- secondary consumers
- primary consumers
- plant
- interconnected
- balance the nature

D Answer the following questions :

- What is a food chain?

- Why are plants called primary producers?

- In the following food chain:

Plants → deer → tiger

If the deer population becomes extinct due to some reasons. What will happen?

CRITICAL THINKING

(Self-Awareness)

Name some animals whose population has gone down due to excessive hunting by man.

LINKING CHAIN

(Writing)

Write a paragraph about 150 words and start from " Hunting and killing animals by man upsets the balance of nature..... "

3

Things Around Us

When we look around, we see different things. They are living as well as non-living.

They have different sizes, shapes and colours. They have weight (mass) and occupy space. All these things are matter.

Everything that has mass (weight) and occupies space is called **matter**.

States of matter



Put into Focus

- ▶ States of matter
- ▶ What is matter made up of?
- ▶ Physical and chemical changes



Solid



Liquid



Gas

Matter is found in three **states** or forms. These states are **solid**, **liquid** and **gas**.

The things in the world are either solid, liquid or gas.

What is matter made up of?

Matter is made up of tiny particles called **molecules**. The molecules are so small that we cannot see them with our naked eyes. The molecules of different substances are different from one another.

Solids

In solids, the molecules are very closely packed. They have a very strong force of attraction among them. They cannot move about freely. Therefore, solids have a definite



Book



Molecules of solid

shape and definite volume. They do not change their shape or the space they occupy. They are generally hard.

Examples– a table, pencil, truck, car etc.

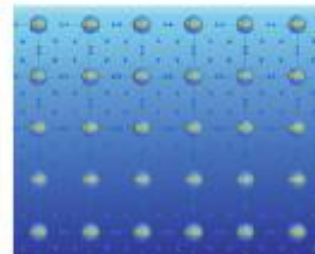
Liquids

The molecules in liquids are less closely packed than solids. They do not have such strong force of attraction between them as in solids.

They can move more freely within the liquid. Therefore, the amount of space that a liquid occupies remains the same, no matter where it is poured. Only the shape changes. This can easily be seen when we pour water from a jug into a glass. The water takes the shape of the glass. The volume of water remains the same.



Water in glass



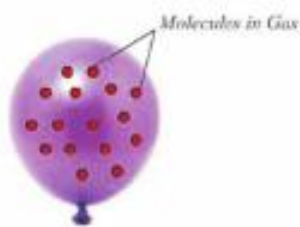
Molecules of liquid

Gases

Molecules in gases are very loosely packed. They have a very weak force of attraction between them. They can move freely. Therefore, gases have no definite shape. They do not have a definite volume. They occupy the entire shape of the container in which they are kept. For example, when a bottle of a perfume is opened in one corner of a room, its fragrance spreads in the entire room.



Balloon



Molecules of gas

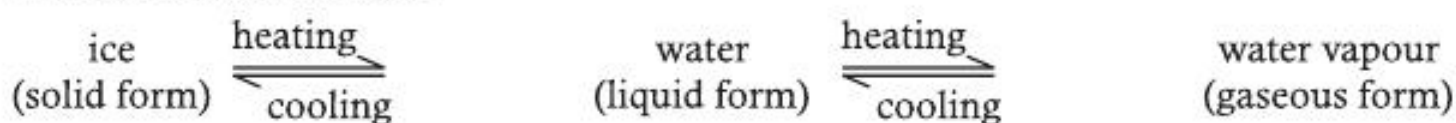
A Comparison

SOLID	LIQUID	GAS
Definite shape	No definite shape. It takes the shape of the container.	No definite shape. It takes the shape of the container.
Definite volume	Definite volume	No definite volume. Its volume is the same as that of the container in which it is kept.
It is generally hard.	Not hard	Not hard
It does not flow.	It flows	It also flows.

States of water are interchangeable



In all cases, the nature of water molecules remains the same. Only the arrangement of water molecules changes.



Physical and chemical changes

Take a piece of paper and cut it into two pieces. It becomes smaller, i.e, its shape changes. But it still remains a paper. The nature of the molecules of the paper does not change. No new substance is formed.



Cutting of paper
(physical change)

A state in which the nature of molecules of the substance does not change is called a physical change. In physical change, no new substance is formed.

Interchangeable nature of water into ice or steam is another example of physical change.

Now burn the pieces of paper. They change into smoke and ash. We cannot get paper back



Burning of paper
(chemical change)

from smoke and ash. So new substances are formed. The molecules of smoke and ash are different from those of paper. Those states in which the nature of the substance changes and new substances are formed, are called chemical changes.



A chemical change

Formation of curd from milk and rusting of iron are the examples of chemical changes.

Solids dissolve in liquids

Take some water in a glass. Add a spoonful of sugar into it and stir it well. The sugar disappears in water.

When you taste this water it is sweet. This is because the sugar has dissolved in water. When a solid such as sugar is dissolved in a liquid like water, it produces a **solution**.

The substances that dissolve in a liquid are said to be **soluble** in that liquid. The solid that dissolves in a liquid is called a **solute**. The liquid in which a substance (solute) dissolves is called a **solvent**. **The liquid we get when a solute is dissolved in a solvent is called a solution.**

In sugar solution, sugar is the solute and water is the solvent. Similarly, in salt solution, salt is the solute and water is the solvent.

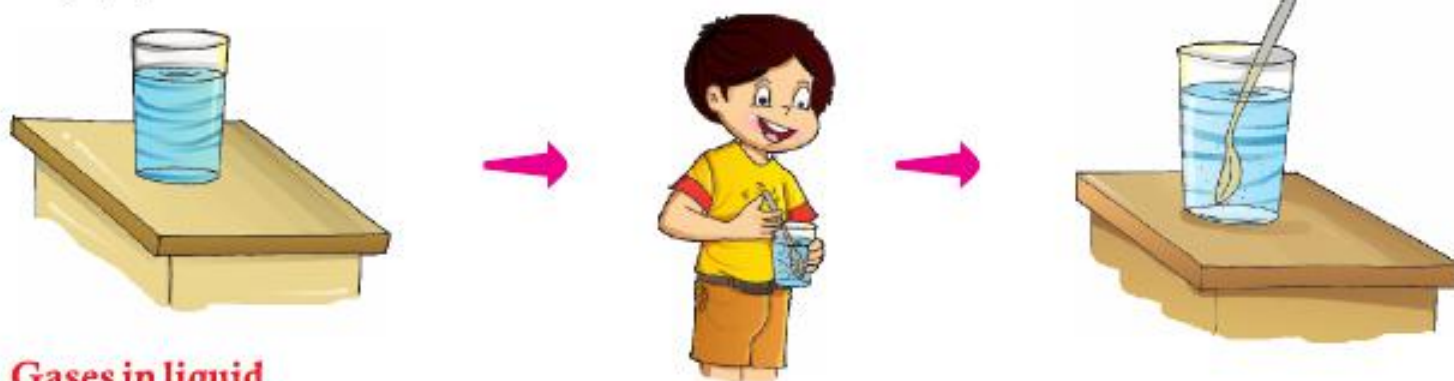


What happens when solids dissolve in liquids?

Take some water in a glass. Mark the water level by pasting a strip of paper outside the glass. Add two teaspoons of sugar to the water and stir it well with a spoon. Note the water level in the glass. It remains the same. This shows that the volume of the solution remains the same as the volume of the solvent.

What does this indicate?

We know that the molecules in a liquid are not very closely packed. There are empty spaces between the molecules. When sugar dissolves in water, sugar molecules fill up the empty spaces between the molecules of water.



Gases in liquid

Many gases like oxygen, nitrogen and carbon dioxide dissolve in water. Carbon dioxide is dissolved under pressure in soda water or any soft drink. When we open the bottle of soda water, we can see that the dissolved carbon dioxide escapes rapidly in the form of bubbles.

The water in lakes, rivers, ponds and seas has a lot of oxygen dissolved in them. Plants and animals that live in water breathe this oxygen dissolved in water.



KEY WORDS

Solid: Firm and stable in shape.

Liquid: A substance such as water or oil that flows freely.

Gas: An airlike fluid substance which expands freely to fill any available space.



OVERVIEW ON THE CHAPTER

- ▶ Matter is found in three states namely solid, liquid and gas.
- ▶ Matter is made up of tiny particles called molecules which can be seen under electronic microscope.
- ▶ Molecules change their position under various situations.

Assignment



COGNITIVE SKILLS



Multiple choice questions :

(Learning by Doing)

Tick (✓) the correct option.

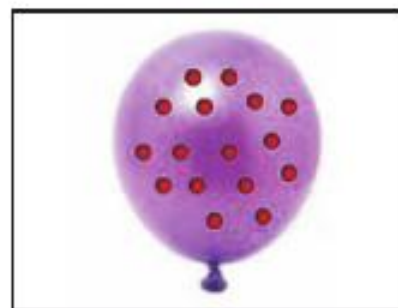
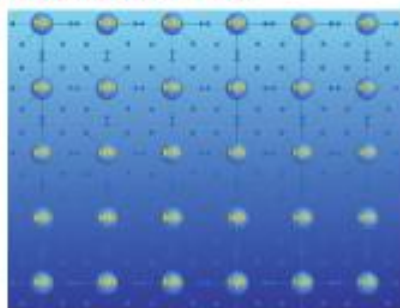
- Matter is made up of tiny particles called _____.
a. molecules ☐ b. solids ☐ c. liquids ☐ d. gases ☐
- A matter that has a definite volume but not definite shape is called _____.
a. solid ☐ b. liquid ☐ c. gas ☐ d. matter ☐
- A matter that cannot flow is _____.
a. gas ☐ b. liquid ☐ c. solid ☐ d. none of these ☐
- Rusting of iron is a _____ change.
a. physical ☐ b. chemical ☐ c. casual ☐ d. none of these ☐



Fill in the blanks :

- Anything that has mass and occupies space is called _____.
- Changing of water into ice is an example of _____ change.
- The liquid we get when a solute is dissolved in a solvent is called _____.
- Plants and animals breathe _____ dissolved in water.

C Identify and name the states of matter by looking at the placement of molecules in the following pictures.



D Answer the following questions :

1. Why do solids have a definite shape?

2. Name any two gases that dissolve in water.

3. Why do liquids flow?

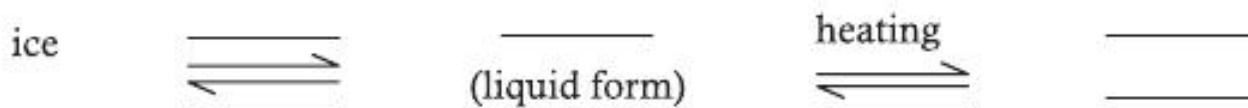
4. List any two differences between solids, liquids and gases.

5. What is the difference between physical and chemical change? Give two examples of each.

CRITICAL THINKING

(Self-Awareness)

Fill in the blanks to complete the following reaction.



PROJECT WORK

(Learning by Experiment)

Make sugar solution with the help of an adult, heat the solution till the water evaporates completely. What is left behind? Take the crystals. What does its taste like?

OBSERVING SKILLS

- See around and write.

(Perceptual Thinking)

Three Solids

Three Liquids

Three Gases

- Circle the odd one out in each group.

- | | | | |
|---------------------|-----------------|------------------|-------------------|
| 1. oxygen | nitrogen | carbon dioxide | solids |
| 2. water | sugar | ice | steam |
| 3. slicing a tomato | rusting of iron | burning of paper | formation of curd |

4

Growing Plants

What is inside a seed?

To learn what is inside a seed and how it grows into a new plant, let us perform the following experiment.

The outer layer of the seed is called the seed coat. It protects the seed. Split the seed and open it gently. What do you see?

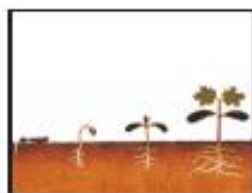
Inside the seed is a baby plant called an **embryo**. Take a magnifying glass and look closer. You will see



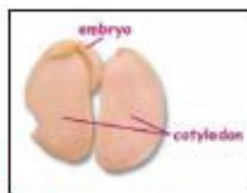
Seed



Seed coat



Seed root



Seed cotyledon

As the stem grows, the plant begins to make its own food. The young plant that grows out from a seed is called a **seedling**.

that the embryo is made up of two parts—two leaves and a root. The leaves store food.

In order to grow, the embryo needs food.

Once the seed germinates, the leaves from the first shoot grow upwards while the root grows downwards. The shoot rises above the ground in search of sunlight

while the roots go down in search of water and minerals.

As the stem



Put into Focus

- ▶ What is inside a seed?
- ▶ Conditions for germination of seeds
- ▶ Dispersal of seeds
- ▶ Ways of seed dispersal
- ▶ Wind dispersal
- ▶ Water dispersal
- ▶ Animal dispersal



Know More!

The formation of a seedling from a seed is known as germination of seed.



Germination

Conditions for germination of seeds

A seed will germinate or sprout, when conditions are right for survival. Usually, a seed has a hard seed coat. It protects the seed from fungi and mould, which grows in presence of moisture. However, when a seed settles in wet soil, it absorbs water and when the weather is warm and sunny, it breaks and grows into a plant.

When a mature seed is placed under favourable conditions and fails to germinate it is said to be **dormant**. The length of time plant seeds remain dormant can be reduced or eliminated by some simple seed treatment called **soaking**, stratification, vernalization.

Dispersal of seeds

Seeds must be **dispersed** over wider area. In this way, they will stand a better chance of getting minerals, water and light, which help the plants to grow well.

Ways of seeds dispersing

Seeds are dispersed in many ways to places far away from their parent plant.

Some of them are as follows :

Wind dispersal

Have you ever seen dandelion seeds float through the air? They look like tiny parachutes.

Some other seeds, like orchid and poppy, are so light that wind easily carries them away.

Some of these seeds have **hair**, which act like parachutes helping the seeds in their flight.



Dandelion



Coconut tree

Water dispersal

Coconut, palm trees grow on beaches. Sometimes, coconut fall into beach and germinate. Sometimes, they fall into the sea, and travel thousands of kilometers before landing on another warm beach, where they may grow into coconut trees.

Animal dispersal

Most plants have juicy fruits that animals like to eat. When animals eat these fruits, only the juicy part is digested while the seeds are excreted. These excreted seeds grow into new plants. This often happens far away from the parent plant.

Explosion

Some plants like the pea plant have pods that explode when



Rabbit



Pea

ripe, the seeds are thrown out when the pod bursts. Those seeds get scattered in different directions and grow as new plants.



Know More!

Velcro which is used widely in sandals, were inspired by idea of spiky seeds.

Human beings can also carry seeds with them to different countries. In fact, many vegetables were brought to India by Britishers and foreign merchants.

Green chillies were brought to India by traders from South America. Seeds of tomatoes, which grow in South America and Mexico originally were transported to England and India.



Water melon

So, you see, we can enjoy a wide variety of fruits, vegetables and flowers only because of the transfer of seeds from one place to another.



KEY WORDS

Active: Energetic.

Embryo: An unborn or unhatched baby or animal in the early stages of development.

Explosion: A violent expansion or bursting with noise.



OVERVIEW ON THE CHAPTER

- ▶ Seeds germinate only when they receive water, air and sunlight.
- ▶ A seed holds a baby plant and a bundle of food.
- ▶ The seed is covered by a seed coat. It contains a baby plant called embryo and cotyledons store food.
- ▶ Formation of young plant from a seed is called germination.
- ▶ It is important for seeds to disperse so that they get better conditions to survive.
- ▶ Seeds are dispersed by wind, water, animals, explosion and humans.
- ▶ Many vegetables, fruits and flowers are brought to India from other countries. Their seeds or seedlings are brought by human beings. This is also a method of seed dispersal.

Assignment



COGNITIVE SKILLS



Multiple choice questions :

(Learning by Doing)

Tick (✓) the correct option.

- A seed is covered by a _____.
a. seed coat ☐ b. embryo ☐ c. cotyledon ☐ d. root ☐
- On _____, the seeds become bigger.
a. seedling ☐ b. soaking ☐ c. drying ☐ d. dispersing ☐
- Young plant that comes out of seed is called _____.
a. seed coat ☐ b. embryo ☐ c. seedling ☐ d. cotyledon ☐
- Scattering of seeds to far-off places is called seed _____.
a. embryo ☐ b. dispersal ☐ c. seedling ☐ d. cotyledons ☐



Match the column A with the column B :

A

B

- | | |
|------------------------|--|
| 1. Germination of seed | a. seeds are dispersed away from their parent plant. |
| 2. Seedling | b. inside a seed, there is a baby plant. |
| 3. Embryo | c. the formation of a seedling from a seed. |
| 4. Dispersal of seeds | d. the young plant that grows out from a seed. |



Describe each of the following in brief :

- A dormant seed
- Embryo
- Seedling
- Germination



Answer the following questions :

- What are the necessary conditions for a seed to germinate?

- What is inside a seed?

- Why is dispersal of seeds important?

- What are the various methods of seed dispersal?

5. Which method of dispersal occurs in the following seeds?

(a) Orchid (b) Coconut (c) Pea plant

CRITICAL THINKING

(Self-Awareness)

1. A seed is protected by a _____.
2. The _____ grows into a new plant.
3. _____ provide food for growth till the seed is planted in the soil.

LINKING CHAIN

(G.K.)

Imagine you are seed then write few lines on this.

ACTIVITY

Make a colourful diagram of seed.

(Creativity)

5

Growing Food

Plants are the major source of food. A large-scale production of food is undertaken to feed people and animals.

This is known as **agriculture**.



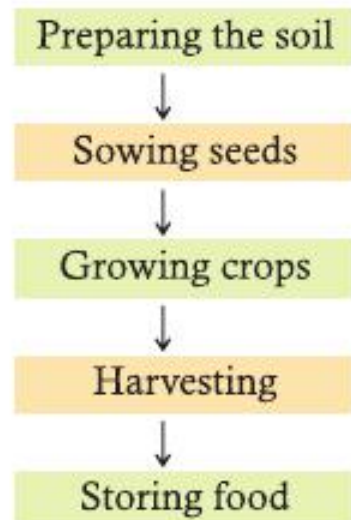
Field

Most fruits that we eat contain seeds.

We also eat seeds of many plants like wheat, maize, legumes, peas etc.

Growing food crops is a process that involves a lot of planning, patience, care and skill.

The following steps are involved in the process of growing food crops.



Preparing the soil

Before seeds are planted, the soil needs to be loosened. This is done with the help of a **plough** and the process is known as **tilling**.

By turning the soil over this way, the soil gets enough air that helps the roots to breathe.



Put into Focus

- ▶ Process involved in production of food
- ▶ Storing food
- ▶ How to grow more food

Sowing seeds

Seeds are generally planted in rows at an appropriate depth. The depth depends on the type of crop. Seeds of plants like legumes (beans, peas, wheat, etc.) are planted 2-2.5cm deep, while corn and potato may be sown at a depth of 6.5-9cm. After placing the seeds, they are covered and gently packed with loose soil.



Sowing Seeds

Growing crops

In the warmth of the sun and water from the rain, the seeds begin to grow into plants. Fertilisers are sprayed on them to help them grow. Sometimes, chemicals are also sprayed on them to get rid of various insects or diseases causing pests. This is how wheat plants look like when they are growing.



Farming crop in field

Harvesting

When the crop is ready, it is cut by hands or with the help of machines. A very useful machine used widely is called a **combine harvester**. It is also known as a **thresher** which separate grains and seeds from straw and chaff.



Harvesting

Storing food



Granary

After harvesting, grains are kept in a store called **granary**.

From the granary, they are sent for **milling**. Milling is the process by which cereal grains are ground into flour. At the mill, it is ground into flour or made into breakfast cereals.

It is important that the food store is free of insects or rats.

Secondly, the store should be completely dry.

Fungi and bacteria, which spoil the food, grow fast in moist condition.

How to grow more food?

The simple answer would be to grow more food. But, it is not so simple because we have limited land. Even today there are millions of people who do not have enough to eat. However, some possible solutions are as follows:



Know More

Sometimes, farmers do not use fertilisers or chemicals. They use manure to provide nutrients to the plants. Manure is made from animal dung and dead leaves. They allow pest controllers like ladybirds, which eat up flies and its larvae, to flourish. Growing onion crop nearly also helps in repelling insects with their strong smell. This is known as organic farming.

- ♦ Watering to improve dry lands. This can be done by building more canals.
- ♦ Use better methods of farming.



KEY WORDS

Plough: A large implement for turning over and cutting furrows in soil.

Tilling: To prepare land for raising crops.

Fertiliser: Any substance used to make the soil more fertile.



OVERVIEW ON THE CHAPTER

- ▶ Most grains, fruits and vegetables are grown from seeds.
- ▶ To produce food on a large scale, crops are grown in farms using the following steps:
1. Preparing the soil 2. Sowing seeds 3. Growing crops 4. Harvesting
- ▶ After harvesting, food is stored in granaries or cold storage. The storage space should be free of insects or pests.



COGNITIVE SKILLS



Multiple choice questions :

(Learning by Doing)

Tick (✓) the correct option.

- The process of turning over or loosening the soil, using a plough is called _____.
a. ploughing ☐ b. tilling ☐ c. harvesting ☐ d. threshing ☐
- _____ and potato are sown at a depth of 6.5-9cm.
a. wheat ☐ b. corn ☐ c. maize ☐ d. rice ☐
- Growing crops without the use of fertilisers and chemicals is called _____ farming.
a. ploughing ☐ b. organic ☐ c. combine ☐ d. tilling ☐
- A place where food grains are stored is called a _____.
a. granary ☐ b. store room ☐ c. cold storage ☐ d. mine ☐



Write T for true and f for false :

1. Fruits and vegetables are stored in a granary.
2. It is important to loosen the soil before sowing seeds.
3. After sowing, the seeds are kept uncovered from the top.
4. It is not possible to grow more food as we have limited space.

☐
☐
☐
☐


Answer the following questions :

1. Name five seeds that we eat.

2. How is soil prepared?

3. What is meant by harvesting?

4. What is a combine harvester?

5. At what depth are seeds generally sown?



Match the column A with the column B :

A

1. After harvesting, grains are kept in a store.
2. Useful machine for harvesting and separation of husk.
3. A large scale production of food is undertaken to feed people and animals.
4. Growing plants with the help of only manure made from animal dung and dead leaves, free of fertilisers and harmful chemicals.
5. Vegetables and fruits are stored in godowns so that, they do not waste.

B

- a. agriculture
- b. cold storage
- c. granary
- d. combine harvester
- e. organic farming

OBSERVING SKILLS

(Perceptual Thinking)

1. Which crop is kept flooded with water to help it to grow?

2. In which country, rice farmers keep fish or duck in rice field?

CRITICAL THINKING

(Self-Awareness)

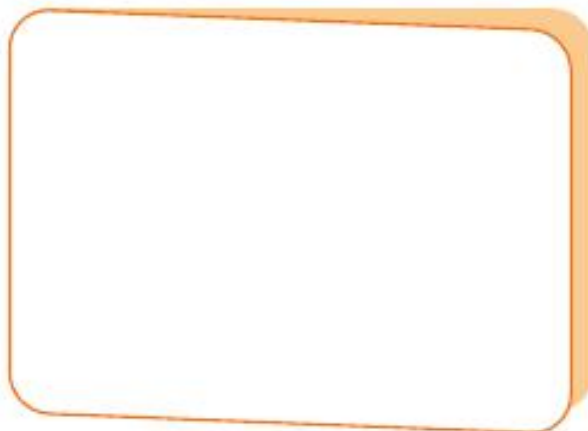
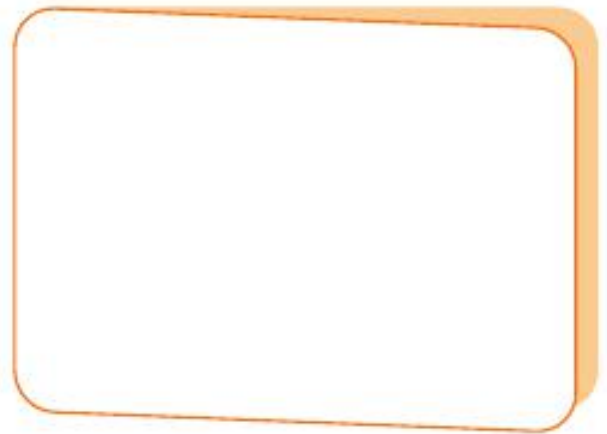
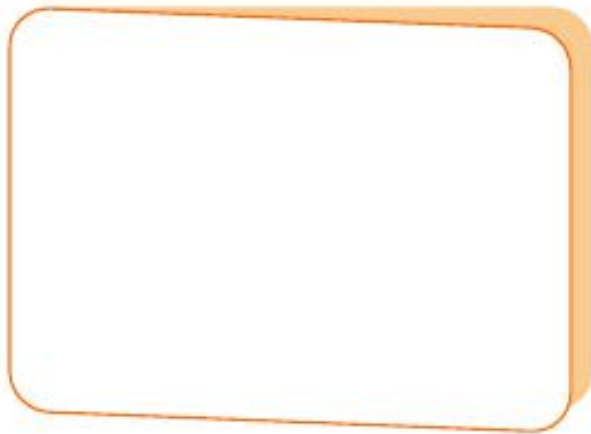
Our country is one of the largest producers of rice. Find out which states of India produce rice.



ACTIVITY

Paste at least three pictures of Indian farmers while working in the space provided.

(Art Integration)



6

Food Preservation

How does food get spoilt?



Spoiled food

Bacteria and **fungi** are present in the air. They start spoiling the food which is few days old. That is why it is very important to eat fresh food. Raw foods like vegetables and fruits should also be properly washed before eating.

They should not be left open for days. When we cook our food, germs like bacteria and fungi present in it, are killed. But if this cooked food is left open it can be attacked again by these germs and get spoilt.

Effect of eating spoilt food

Effects of eating spoilt food can cause diseases like dysentery, typhoid, cholera and even death.

Spoilt food also causes food poisoning.

Food poisoning can cause abdominal pain, diarrhoea, vomiting and fever.

Food preservation

Food preservation is the process of treating and handling food to stop spoilage by bacteria and fungi.

Food can be prevented from going bad by avoiding the conditions under which bacteria and fungi grow.

Bacteria and fungi are living things. They need air, water and warmth to live. That is why we keep food in a refrigerator to stop it from going bad.

Bacteria and fungi cannot grow inside a cold refrigerator.

These are the two ways in which we preserve our food :

Dry storage Cold storage



Put into Focus

- ▶ How does food get spoilt?
- ▶ Effects of eating spoilt food
- ▶ Food preservation
- ▶ Wastage of food

Dry storage

The storing of non-perishable food items, that does not go bad easily like sugar, rice, wheat, nuts and pulses in a dry place is known as **dry storage**. It is one of the oldest methods of food preservation.

This process prevents water from entering food and this stops bacterial or fungal growth. The process is often applied to preserve fruits like apple, pears, bananas, mangoes, papayas, apricots and coconuts.



Apple



Banana



Mango



Papaya



Apricot



Coconut

Cold storage

Storing of perishable food items that go bad easily like jam, jelly, pickle and cooked dishes in a cold place is known as **cold storage**.



Food in refrigerator

The refrigerator is a cold storage.

Methods of food preservation

Freezing

Bacteria and other germs cannot grow in the cold. Thus, food are frozen to stop them going bad. But once, the food is taken out of the refrigerator, it must be eaten immediately as it can again get spoilt and make us sick.



Know More

Meat and fish both can be preserved and flavoured through the use of smoke. This is known as smoking.

Drying

Water in a food item dries up if it is kept in the sun. Bacteria and fungi and other germs cannot live without water. Thus, dry food does not go bad.

This is also a traditional method of preserving fruits and fish.

Adding preservatives

Preservatives are chemicals that are added to different food items to stop them from going to bad.

Salt is the most commonly used preservative. It removes water from the food and stops germs from growing. Other common food preservatives are vinegar and sugar.

Pickles are made with vinegar. Green chillies and cucumber are preserved with vinegar, salt and sugar.

Jams and squashes are also preserved with the help of preservatives.



Pickle



Jam



Squash

Heating and boiling

Bacteria and fungi cannot live in high temperature. Thus, if foods are heated and boiled, all the germs in them are killed.

That is why milk is boiled.

Packing

Food can also be preserved by packing them in polythene bags. Germs cannot grow without air.

Polythene bags do not allow any air to go in.

So, cheese and cooked meat are packed in this way.

Canning

Food industries use this process for preservation.

Food is first heated to kill germs and then stored in clean containers or cans.

Foodstuffs preserved by this process can be kept inside the refrigerator as long as they are not opened. Canning



Canning

preserves food for a long time but not forever. We should always check expiry date printed on the cans before buying such food.

They get spoilt after the expiry date.

Wastage of food

Nearly 25,000 people in the world die of starvation everyday. We must buy only as many food items as we need. We must cook only as much as we can eat and preserve. We must eat only as much as we need.

We should store and preserve food properly to prevent wastage.



KEY WORDS

Bacteria: Bacteria are very small organisms.

Fungi: An organism without leaves or flowers.

Preserve: Keep in its original or existing state.



OVERVIEW ON THE CHAPTER

- ▶ Bacteria and fungi spoil food.
- ▶ Bacteria and fungi are present in air.
- ▶ Food can be preserved by dry storage and cold storage methods.
- ▶ Food can be preserved by drying, sunning, freezing and adding preservatives.
- ▶ Bacteria and fungi cannot live without air or in very hot or cold temperature.
- ▶ We should not waste food.



Assignment



COGNITIVE SKILLS



Multiple choice questions :

(Learning by Doing)

Tick (✓) the correct option.

1. _____ preserves food for a long time.

a. freezing

c. packing

b. canning

d. drying

2. Food preservation means _____.
a. food storage ☐ b. food wastage
c. food spoilage ☐ d. none of the above
3. Bacteria cannot live without _____.
a. air ☐ b. water
c. fungi ☐ d. none of the above



Fill in the blanks :

- _____ and _____ spoil food.
- _____ is a traditional way of preserving food.
- _____ is one of the oldest ways of preserving food.
- _____ is the most commonly used preservative.
- _____ and _____ are common preservatives.



Match the column A with the column B :

A

- Apple
- Mango
- Milk
- Fish
- Meat

B

- smoke
- pasturisation
- drying
- jam
- pickle



Answer the following questions :

- How does food get spoilt?

- How do we know that a food item is spoilt?

- What happens when we eat spoilt food?

- What are the different processes of food preservation?

- How can we stop the wastage of food?

OBSERVING SKILLS

(Perceptual Thinking)

Look at these food items and decide which method of preservation you will use to store them.

1. mangoes _____

2. fishes _____

3. oranges _____

4. cooked rice _____

5. peas _____

6. squash _____

ACTIVITY

(Creativity)

Find out about traditional Indian methods of making pickles. Paste the pictures of making pickles.



7

Not Enough Food

Our body is like a machine. Like all machines, we need energy to work. From where do we get this energy?

We get this energy from the food we eat.

Famine



Famine

In some parts of the world, people do not get enough food to eat because there is a severe shortage of food in their country.

Quite a few of them die due to starvation or suffer from deficiency diseases.

The condition when people do not get enough food to eat over a long period of time, is called **famine**.

Famines are caused by:

- ◆ Natural calamities
- ◆ Man - created problems

Natural calamities

Water plays an important role in crop production.

Most of the farmers depend on rainwater for their crops. When there is little or no rain, no crops grow or we say there is failure of crops. This leads to a shortage of food in the area. Again,



Earthquake

when there is excessive rainfall, crops get destroyed. This again leads to a shortage of food.

Natural calamities like **earthquakes**, **plant diseases** and volcanic eruptions also damage crops and lead to shortage of food.



Put into Focus

- ▶ Famine
- ▶ Natural calamities
- ▶ Man-created problems
- ▶ Deficiency diseases



Flood



Volcanic eruption

Man-created problems

Improper food storage : Storage of food grains should be done in **godowns** that are free of micro-organisms, insects and rats. Food grains that are not stored properly get destroyed.

This destruction of grains also leads to a shortage of food.

Poverty : Poor people do not have enough money to buy food. Some of them get only one meal a day. So, they suffer from **malnutrition** and **starvation**.

Hoarding and black marketing : In order to earn more profits, some shopkeepers hide and store food grains in their godowns. This is called **hoarding**. Later they sell the food grains at a very high price in the black market. Hoarding causes a shortage of food at the market. Poor people cannot purchase food grains at high prices. So, there is a shortage of food. Both hoarding and black marketing are **illegal** and **punishable by law**.

Deficiency diseases

People who do not get enough food to eat suffer from **malnutrition**. Malnutrition leads to **deficiency diseases**.

Diseases that are caused by lack of nutrients such as carbohydrates and vitamins in our diet are called **deficiency diseases**.

Deficiency disease	Caused by lack of this nutrient	Symptom	Food giving deficient nutrient
1. Marasmus 	Carbohydrates	The child becomes too thin that loose folds of skin can be seen all over the body. The ribs of the child can be seen.	Rice, wheat, potatoes, maize, honey
2. Kwashiorkor 	Proteins	The abdomen of the child gets swollen and bulges out. The legs become thin.	Eggs, fish, meat, cereals, pulses
3. Anaemia 	Iron	The blood carries less oxygen in the body resulting in pale looks, tiredness, loss of appetite and weight.	Eggs, pulses, green leafy vegetables, spinach, cereals

4. Goitre		Iodine	Swelling of the thyroid gland.	Fish, sea food, iodised salt
5. Improper development of bones and teeth		Calcium and phosphorus	Bones and teeth become weak and brittle.	Milk, green vegetables, meat and fish, pulses, nuts
6. Rickets		Vitamin D	Bow legs (bent legs), bending of the spine.	Milk, fish, eggs
7. Night blindness		Vitamin A	The person cannot see in dim light.	Carrot, butter, milk, spinach, green vegetable.
8. Beri Beri		Vitamin B	Skin becomes dry and scaly, extreme weakness, the nervous system and the muscles are affected.	Milk, eggs, meat, green leafy vegetables
9. Scurvy		Vitamin C	Swelling and bleeding of gums.	Citrus fruits e.g lemon, oranges and tomatoes



KEY WORDS

Earthquake: A sudden violent movement in the earth's crust.

Hoarding: To collect and often hide away a supply of something.

Malnutrition: Poor health caused by not having enough food.



OVERVIEW ON THE CHAPTER

- ▶ The condition when people do not get enough food to eat over a long period of time called famine.
- ▶ Famines are caused by natural calamities and man-created problems.
- ▶ Diseases which are caused by lack of nutrients called deficiency diseases.
- ▶ Some of the deficiency diseases are Marasmus, Kwashiorkor, Anaemia, Goitre.

Assignment

COGNITIVE SKILLS



Multiple choice questions :

(Learning by Doing)

Tick (✓) the correct option.

1. _____ plays the most important role in crop production.
 a. water ☐ b. food ☐ c. shelter ☐ d. air ☐
2. Malnutrition leads to _____.
 a. fitness ☐ b. deficiency ☐
 c. black marketing ☐ d. none of these ☐
3. Both hoarding and black marketing are _____.
 a. legal ☐ b. illegal ☐
 c. starvation ☐ d. none of these ☐
4. When the abdomen of a child gets swollen and bulges out, he is suffering from _____.
 a. marasmus ☐ b. anaemia ☐ c. kwashiorkor ☐ d. rickets ☐



Fill in the blanks :

1. Most of the farmers depend on _____ water for their crops.
2. A person suffering from _____ cannot see in dim light.
3. In scurvy, there is _____ and _____ of gums.
4. Diseases caused due to lack of nutrients are _____ diseases.



Answer the following questions (short) :

1. What is meant by famine?

2. What are natural causes of famine?

3. What is meant by hoarding? Why is it illegal?
- _____



Answer the following questions (Long) :

1. How does improper food storage lead to shortage of food?
- _____
- _____

2. What are deficiency diseases? Name the diseases caused by the deficiency of the following nutrients.

a. carbohydrates

c. vitamin A

e. iodine

g. calcium

b. iron

d. vitamin B

f. vitamin C

h. protein

3. Name two foods for each that we should eat to avoid the following diseases :

1. Marasmus :

2. Goitre :

3. Night blindness :

OBSERVING SKILLS

Circle the odd one in each group. Give the reason for your answer.

(Perceptual Thinking)

1. flood

earthquake

volcanic eruption

hoarding

2. marasmus

kwashiorkor

malaria

goitre

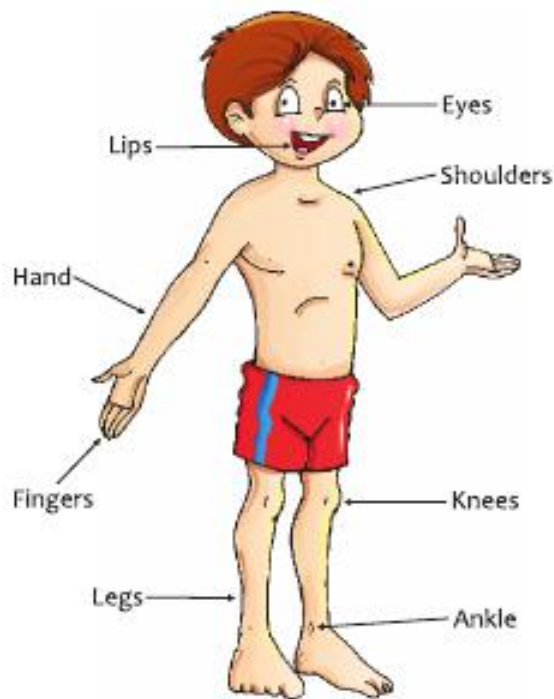
COMMUNICATION SKILLS

(Discussion)

Do group discussion on the topic "Red cross and many organisations. Help people during Natural Calamities".

8

Organ System



Boy

The building blocks of our body are called **cells**.

These cells join together to form **tissues**.

Similar tissues combine to form organs and organs work together to do a similar kind of work. They form an **organ system**.

A group of organs working together to carry out one major function of the body is called a **system**.

The main systems of body are :

- ◆ Skeletal system
- ◆ Nervous system
- ◆ Digestive system
- ◆ Excretory system
- ◆ Muscular system
- ◆ Respiratory system
- ◆ Circulatory system
- ◆ Reproductive system



Put into Focus

▶ Human body systems

Skeletal system

Our body is made up of **206 bones**. They are very hard and join together to make up a framework. The bony framework of our body is called the **skeletal system**. We are able to stand straight and walk due to our skeleton.

It gives shape and support to our body. It also protects our soft organs like brain, heart, lungs etc.

Muscular system

The soft and fleshy parts of our body are muscles and most of them are fixed to the bones.

Muscles along with bones help the joints to move. In fact, every movement of our body is controlled by the



Skeletal system



Muscular system

muscles.

There are more than 600 muscles in our body.

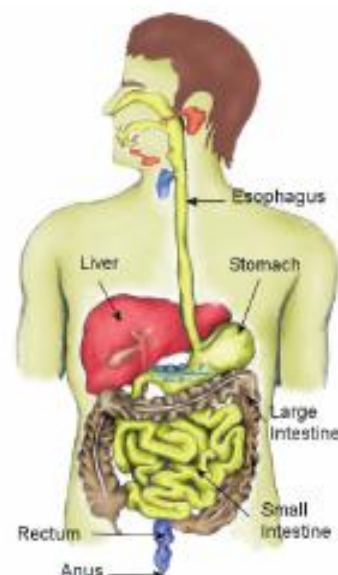
Digestive system

The food we eat is complex in form.

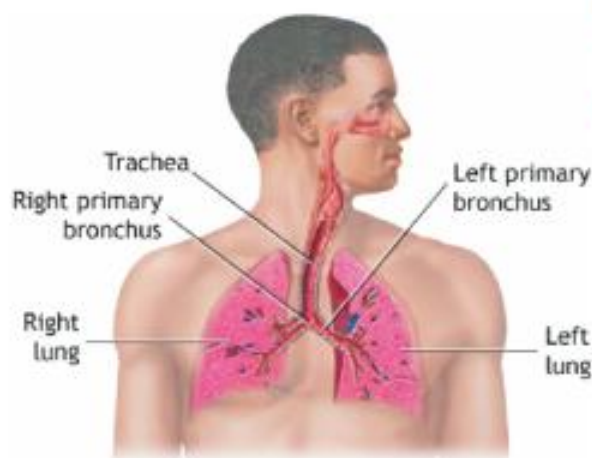
So, this source of energy is broken down into small pieces and then absorbed by the body.

This process is called **digestion**. The organs that help in digestion form the **digestive system**.

Our teeth breakdown the food into simpler form. Then these small pieces goes to the **stomach** through the food **pipe**. Digestive juices of the **stomach**, mix with food and **food** form a paste. And now this food goes into **small intestine** where its juices further digest the food. The useful part of the food is then absorbed into the blood. The rest of the food that cannot be digested by our body goes into **large intestine**. It is finally thrown out of the body through **anus**.



Digestive system



Respiratory system

Respiratory system

Breathing in and breathing out is called **respiration**. The respiratory system consists of nose, windpipe and lungs.

When we breathe in, air enters through our nose and goes down into the windpipe and then finally into the lungs. The lungs absorb oxygen from the air we breathe in. This oxygen burns the food, purifies the blood and gives energy and carbon dioxide. When we

breathe out, carbon dioxide comes out of lungs.

Nervous system

Our brain is the control station or engine of our body.

It controls all the organs of the body. All sense organs send messages to the brain through nerves. **Brain**, **spinal cord** and **nerves**

make up the nervous system. The brain controls the different body parts through these nerves.



Nervous system



Know More!

Average weight of human brain is approx 1.4 kg.

Circulatory system

The **heart**, **blood** and blood vessels make up the circulatory system. The heart is located in our chest and many tubes come out of heart. These are called **blood vessels**.

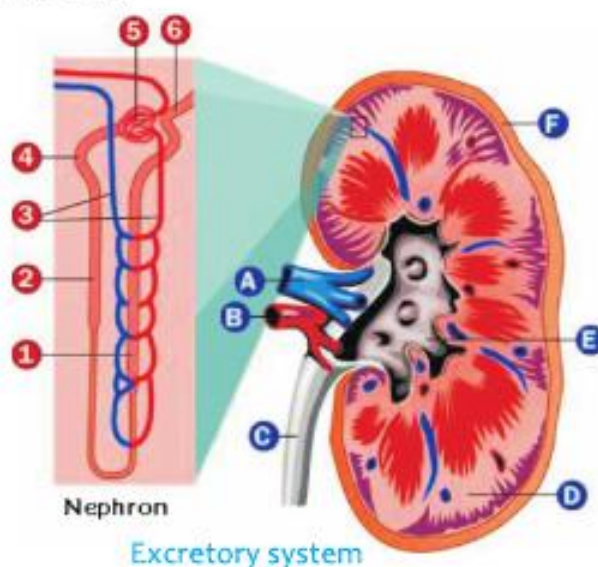
The heart pumps blood into these blood vessels.

These blood vessels carry the blood to all parts of the body.

Excretory system

Our body produces a lot of waste material, which is harmful for the body. So, it has to be removed for the proper working of the body. This is done with the help of **kidneys** and **skin**. They form the excretory system.

Two bean shaped kidneys help to keep the blood clean and filter out urine. Carbon dioxide is removed by lungs. Extra salt and water are removed by the skin in the form of sweat.



Reproductive system

Reproduction is very important and necessary system to maintain the life cycle.

There are special organs in the body which are responsible for producing babies.



KEY WORDS

Muscles: A band of bodily tissue that can contract to move a body part.

Digestion: Breakdown food in the body so that it can be easily absorbed.

Respiration: The act of breathing.



OVERVIEW ON THE CHAPTER

- ▶ Cells are building blocks of our body.
- ▶ A group of organs work together and form an organ system.
- ▶ Different parts of body have different works to do.
- ▶ The brain controls the different organs of the body.
- ▶ Various systems like nervous system, digestive system, etc. are responsible for the proper functioning of the body.

Assignment

COGNITIVE SKILLS



Multiple choice questions :

(Learning by Doing)

Tick (✓) the correct option.

- System that carries blood to all parts of the body is _____.
 a. circulatory system ☐ b. respiratory system ☐
 c. digestive system ☐ d. reproductive system ☐
- _____ are the building blocks of body.
 a. organs ☐ b. tissues ☐ c. cells ☐ d. muscles ☐
- Air enters the lungs through _____.
 a. mouth ☐ b. windpipe ☐ c. nostrils ☐ d. muscles ☐
- The system removes waste from our body is _____.
 a. excretory system ☐ b. respiratory system ☐
 c. digestive system ☐ d. reproductive system ☐
- The brain sends messages to all parts of the body through _____.
 a. blood ☐ b. nerves ☐ c. foodpipe ☐ d. nostrils ☐



Match the column A with the column B :

A

- Kidneys
- Heart
- Lungs
- Stomach
- Brain

B

- digestion
- respiration
- control body activities
- excretion
- circulation



Fill in the blanks :

- The tubes in the body that carry blood are called _____.
- The process of breathing in and out is called _____.
- _____ are the building blocks of life.
- _____ system supplies food and oxygen to all parts of the body.
- _____ system gives shape and support to human body.



Write 'T' for true and 'F' for false :

1. Brain is the controlling centre of our body.
2. Heart, blood and blood vessels make up the digestive system.
3. Excretory system removes waste from our body.
4. The lungs help us to breathe.
5. The food we need is taken in by the digestive system.

☐
☐
☐
☐
☐


Answer the following questions :

1. Define an organ system.

2. How do muscles help us?

3. What are the main organs of circulatory system?

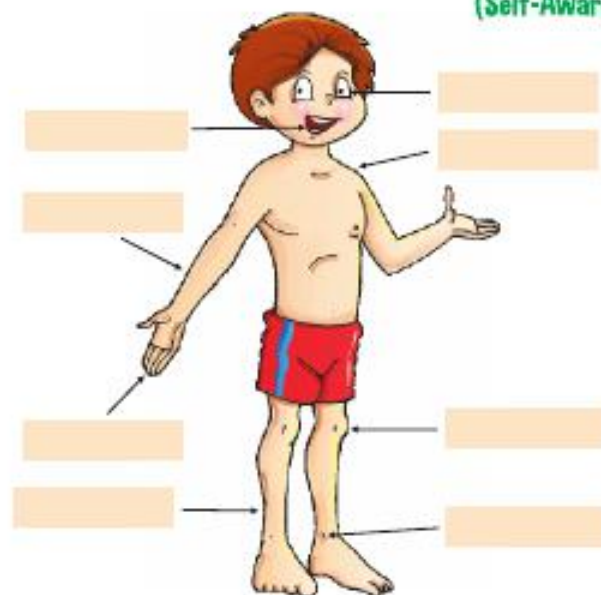
4. Name the organs of the respiratory system.

5. What is the function of kidneys?

CRITICAL THINKING

Label the diagram showing different body organs.

- | | | |
|--------|-----------|-------------|
| → lips | → fingers | → hands |
| → eyes | → legs | → shoulders |
| | → knees | |
| | → ankle | |



(Self-Awareness)

OBSERVING SKILLS

(Perceptual Thinking)

1. Press your knees and elbows, are they hard or soft?
2. Take a deep breathe. When you breathe in, your chest expands. Now breathe out slowly. What happens to the size of chest?

9

Reproduction in Animals

All living things reproduce. Some animals lay eggs, while others give birth to their young ones. **The process by which animals produce their young ones is called reproduction.**



Put into Focus

- ▶ Animals that give birth to young ones
- ▶ Animals that lay eggs

Animals that give birth to young ones



Deer with its fawn

Animals that give birth to young ones and produce milk to feed them are called **mammals**. These animals carry their young ones inside their bodies till they born.

Dogs, cats and elephants are mammals.

Human beings are also mammals. **Whales** and **dolphins** look like fish but they are also mammals. Mammals are the most developed amongst animals.

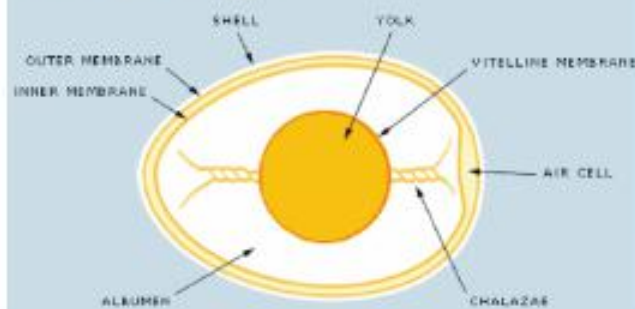
Animals that lay eggs

Birds, fishes, insects, frogs, snakes and crocodiles are the **animals that lay eggs**.



Cow with its calf

Anatomy of an Egg



Internal structure of an egg

Birds

All birds lay eggs. Female birds lay eggs in their nests. Each egg has a hard protective outer shell. Inside the shell, there is a clear sticky liquid called **albumen** (also called egg white).

Inside it, there is a yellow liquid called **yolk** that **provides** food to the baby growing inside.

This baby is called the **embryo**. The embryo develops into a chick only when egg is kept

warm. The parent bird keeps the egg warm by sitting on it.

After 21 days, the fully grown chick breaks egg and comes out. This is called **hatching**.

The young ones do not have feathers and their eyes are closed. The parent birds feed them and teach them how to fly. They look after them till they are strong enough to look after themselves.



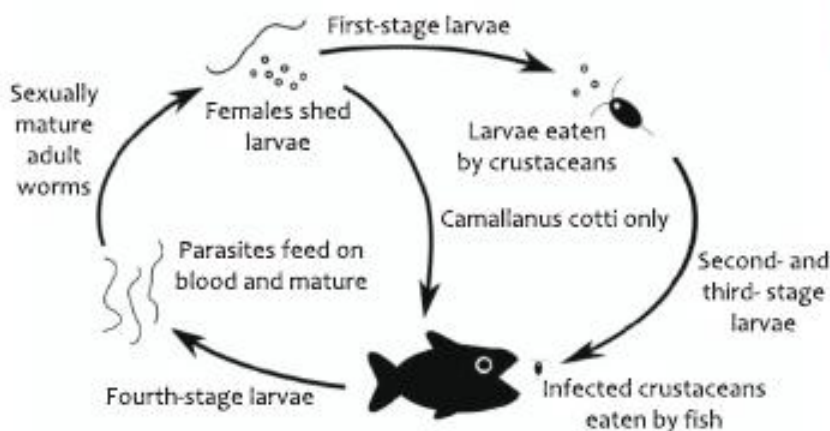
Chick hatching out an egg

Fishes

Fishes lay thousands of eggs in safe places in water. The eggs of a fish are called **spawn**.

A young baby fish hatches and grown into a **fry**.

The fry grows into an adult fish. Out of the thousands of eggs the fish lays, only a few hundreds grow into baby fish. Many eggs are eaten by other big fish.



Life cycle of fish

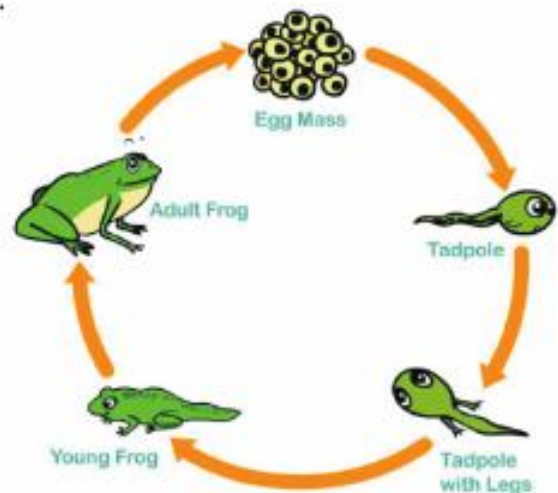
Frogs

A frog lays eggs in a safe place in water, like the fish.

A frog's eggs are also called **spawn**. About three weeks later, baby frogs come out of the eggs. They are called **tadpoles**. They look like tiny fish and breathe through **gills**.

The tadpoles gradually lose their tails and develop legs. After sometime, they become adult frogs.

Like fish, parent frogs take care of their babies only for a short time.



Life cycle of frog



Crocodile

Reptiles

Reptiles like lizards, turtles and snakes lay their eggs on the ground. They do not sit on the eggs to keep them warm. The eggs are warmed by the heat of the sun.

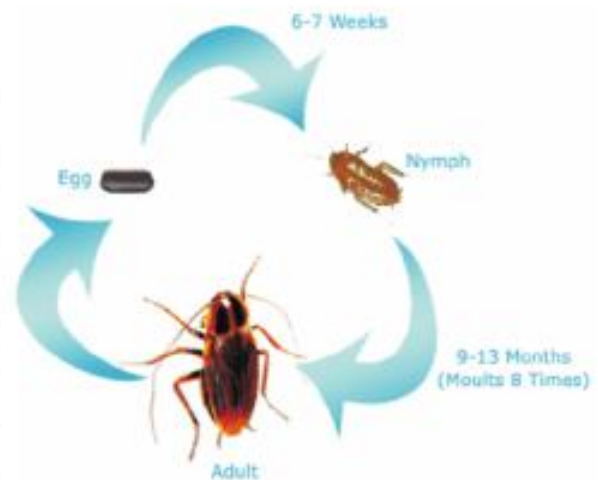
Crocodiles dig shallow pits in sandy river banks to lay their eggs. The mother covers the eggs with sand and guards them. When the babies hatch out from the eggs, they dig

away the sand and sets them free. As the young ones come out, the mother keeps them in a special pouch at the bottom of her mouth for sometime.

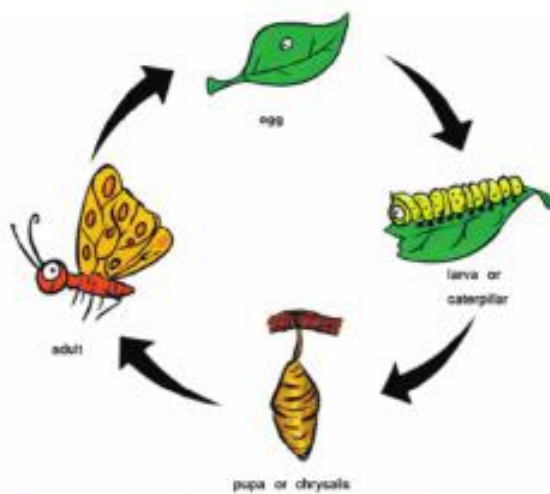
Insects

All insects lay eggs. Insects like grasshoppers and cockroaches have three stages in their life cycle. The baby cockroach is called a **nymph**. It sheds its skin several times and changes into an adult. Shedding of old skin is called **moulting**.

Other insects like butterfly and housefly have four stages in their life cycle. The female butterfly lays eggs in clusters on the underside of the leaf. Each egg hatches into a **larva** that looks like a worm. This is the **caterpillar**. It eats leaves and grows quickly.



Life cycle of cockroach



Life cycle of butterfly

After sometime, the caterpillar stops eating and forms a shell called **cocoon** around its body. This stage is called **pupa**. Inside the pupa, the caterpillar slowly changes its shape. After a while the pupa bursts and an adult butterfly comes out.



Know More!

Birds lay eggs in nests. **Turtles** bury their eggs in the ground. **Cockroaches** hide their eggs in dark corners.



KEY WORDS

Reproduce: Give birth to young ones of the same kind.

Mammal: A warm blooded animal that produces milk and bears live young.

Albumen: Egg white.



OVERVIEW ON THE CHAPTER

- ▶ All living things reproduce.
- ▶ Animals that give birth to young ones and produce milk to feed them are called mammals.
- ▶ All birds lay eggs.
- ▶ Fishes lay thousands of eggs in water.
- ▶ Reptiles lay eggs on the ground.

Assignment



COGNITIVE SKILLS



Multiple choice questions :

(Learning by Doing)

Tick (✓) the correct option.

- The yellow substance inside a bird's egg is called _____.
a. albumen ☐ b. embryo ☐ c. yolk ☐ d. moulting ☐
- The baby frog that comes out of an egg is called _____.
a. tadpole ☐ b. fry ☐ c. nymph ☐ d. spawn ☐
- The larva of a butterfly is called _____.
a. cocoon ☐ b. caterpillar ☐ c. pupa ☐ d. yolk ☐
- Tadpoles breathe through their _____.
a. lungs ☐ b. gills ☐ c. skin ☐ d. kidneys ☐



Fill in the blanks :

- _____ looks like fish but is a mammal.
- A baby cockroach is called a _____.
- Shedding of old skin in insect is called _____.
- A frog's eggs are also called _____.



Write the names of the following pictures in the space provided :





Answer the following questions :

- Why do birds sit on their eggs?

- Why do fish lay thousands of eggs in water?

- Name the four stages in the life cycle of butterfly.

- Draw a diagram to show the three stages in the life cycle of a cockroach.

CRITICAL THINKING

(Self-Awareness)

Name any three animals that can

- a. Give birth to young ones : _____
- b. Lay eggs : _____

PROJECT WORK

(Learning by Experiment)

Observe a chick hatching out of an egg. Also observe its features. Record it your project file.

ACTIVITY

Paste the pictures of bird which can able to fly very fast. Paste at least six pictures. (Art Integration)



10 Teeth and Tongue

Our teeth are a very important part of our body.

1. They give proper shape to our face.
2. They help us in biting and chewing food.
3. They help us to speak clearly.

There are many words that we cannot say without our teeth.

Changing teeth

A newly born baby does not have any teeth.

When the baby is six to seven months old, teeth begin to appear. By the age of three, the child has a set of 20 teeth. This set of teeth is called **milk teeth** or **temporary teeth**.

When the child is six years old, it starts losing these milk teeth and new teeth grow in their place. This new set of teeth is called **permanent teeth**.

By the time, the child is 12 to 13 years old, all permanent teeth grow. Generally there are 28 permanent teeth. The last four teeth are known as **wisdom teeth**. These teeth come out between the ages of 20 and 30. Therefore, adults have a set of 32 teeth -16 in the upper jaw and 16 in the lower jaw. When we lose any teeth of this set, we do not get a new one in its place.

Structure of a tooth

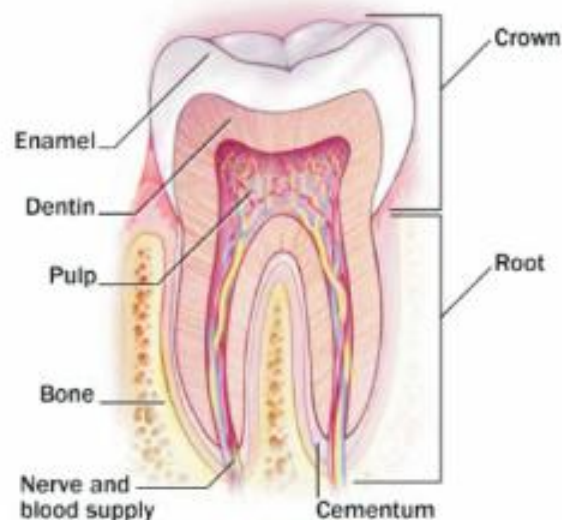
When we open our mouth and look in a mirror, we see our teeth. Teeth are hard, strong and bony structures fixed firmly in the upper and lower jaws. Our teeth grow out of our soft pink **gums**. Gums hold our teeth. Only a portion of the teeth can be seen.

The portion of the tooth seen above the gum is called the **crown**. The portion of the tooth inside the gum which we cannot see, is called **root**.



Put into Focus

- ▶ Changing teeth
- ▶ Structure of a tooth
- ▶ Kinds of teeth
- ▶ Tooth decay
- ▶ Care of teeth
- ▶ Our tongue



crown and root

A tooth has an outside white covering called **enamel**. This is the hardest material in the human body. The next layer inside the enamel and forming the main part of the tooth is called **dentine**.

The centre of each tooth is called **pulp**. The pulp is soft and has blood vessels and nerves in it. These blood vessels provide food to the teeth.



Structure of teeth

Kinds of teeth



Incisor



Canine



Premolar



Molar

When we look at our teeth in the mirror, we see that they are of different sizes and shapes. There are four kinds of teeth.

1. Incisors : These are four front teeth in each jaw. They have sharp and straight edges. They are used to cut and bite food. They are also called **cutting teeth**.

2. Canines : There are two canines in each jaw. They are present on either side of the incisors. They are very sharp and pointed like a fork. They help in tearing food. For this reason, these are called **tearing teeth**.

3. Premolars : There are four premolars in each jaw. They are next to canines. They are broad and flat teeth. These teeth help us cracking food. They are called **cracking teeth**.

4. Molars : There are 6 molars in each jaw. They are next to premolars. They are broader than premolars. They help in grinding the food. They are also called **grinding teeth**.

Tooth decay

When we eat, small bits of food get stuck between our teeth. Sometimes, these small bits of food are not removed when we do not brush our teeth carefully. This stuck food rots and our teeth get covered with a thin sticky layer of germs. This sticky layer of germs is called **plaque**. This plaque reacts with the sugar present in the food to produce **acid**. The acid affects the tooth enamel and starts destroying it. This is called **tooth decay**. It causes holes or cavities in the enamel. As a result the nerves and the blood vessels in the pulp get exposed. When any cold or hot thing comes in contact with these exposed nerves, we feel pain or toothache. Thus, tooth decay causes the following :

1. Toothache
2. Foul smell from the mouth
3. Stomach disorders
4. Holes or cavities in the teeth
5. Loss of teeth

Care of teeth

Teeth are very important part of our body. We must take care of our teeth to prevent tooth decay.

1. We must brush our teeth at least twice a day, in the morning and at night before going to bed.
2. We should brush our teeth by following correct methods as shown below.
 - a. Outside surface of upper teeth use a downward sweeping motion.



Brushing on the inside surface of upper teeth

- b. Inside surface of upper teeth use a downward sweeping motion.



Brushing on the upper teeth

- c. Inside surface of lower teeth use an upward sweeping motion.



brushing on upward sweeping motion

- d. Surface of premolars and molars scrub back and front.



Healthy food

3. We should eat raw vegetables and fruits like apples, carrots and radish. They make our teeth and gums strong.
4. We should avoid sweets and chocolates. They spoil our teeth.



Sweets and chocolates

5. We should visit a dentist every six months.
6. Teeth that are not kept clean get covered with a yellow substance called **tartar**. This is removed by a dentist.
7. We must floss our teeth. A **dental floss** is a piece of string that is moved back and forth through the spaces between our teeth. Flossing removes the food that cannot be removed by brushing.



Flossing



Wire braces



Know More

When teeth grow crooked, the dentist fits wire braces to straighten them. Wire braces help the teeth to grow straight.

Our tongue

Our tongue is an important taste organ. There are four kinds of taste buds- sweet, sour, bitter and salty present on our tongue. These help us to taste the food.

Try to hold your tongue and speak. You find it very difficult. Stand in front of a mirror and speak. You see your tongue roll when you speak. This rolling action of the tongue helps you to pronounce the words correctly.

The tongue, therefore plays an important part in speaking.



KEY WORDS

Dentine: The hard tissue family, the main part of a tooth.

Pulp: A soft, wet, mass of crushed material.

Tartar: A hard deposit on teeth.



OVERVIEW ON THE CHAPTER

- ▶ Teeth plays a very important part in our body.
- ▶ There are milk teeth, permanent teeth and wisdom teeth.
- ▶ Kinds of teeth includes: - Incisors, Canines, Premolars and Molars.
- ▶ We must look after our teeth carefully.

Assignment

COGNITIVE SKILLS



Multiple choice questions :

(Learning by Doing)

Tick (✓) the correct option.

- The last four teeth that come out between the ages of 20 and 30.

a. incisors	☐	b. canines	☐
c. wisdom teeth	☐	d. permanent teeth	☐
- The centre part of each tooth is _____.

a. dentine	☐	b. pulp	☐	c. crown	☐	d. root	☐
------------	-----------------------	---------	-----------------------	----------	-----------------------	---------	-----------------------
- Molars are also called _____ teeth.

a. grinding	☐	b. cutting	☐	c. tearing	☐	d. none of these	☐
-------------	-----------------------	------------	-----------------------	------------	-----------------------	------------------	-----------------------
- Uncleaned teeth get covered with yellow substance called _____.

a. tartar	☐	b. holes	☐
c. tooth decay	☐	d. dentine	☐



Fill in the blanks :

- The set of teeth a three-year-old child has is called _____.
- _____ hold our teeth.
- The _____ is the sense organ of taste.
- _____ removes the food that cannot be removed by brushing.



Name the following teeth :





Answer the following questions :

- Why are teeth important?

- Why should we avoid chocolates and sweets?

3. What is tartar?

4. What is a dental floss?

5. What are the harmful effects of tooth decay?

OBSERVING SKILLS

Circle the odd one in each group.

(Perceptual Thinking)

- | | | | |
|---------------|---------|---------|--------|
| 1. incisors | yeast | canines | molars |
| 2. enamel | dentine | pulp | tongue |
| 3. ice-creams | carrot | radish | apple |

COMMUNICATION SKILLS

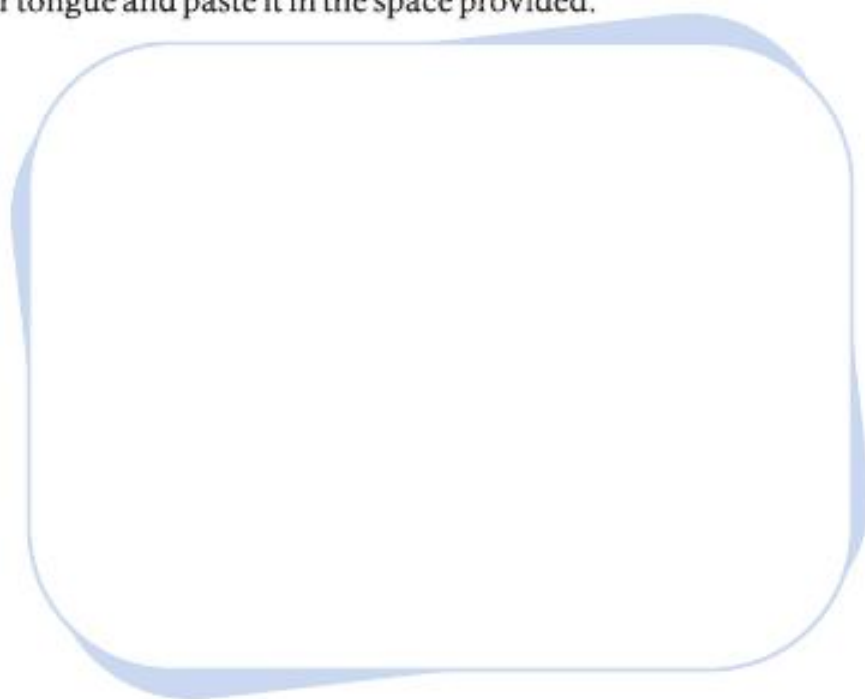
(Discussion)

Write down ten words that you can say properly and ten words that cannot say without the tongue touching your teeth.

ACTIVITY

Take a picture of your tongue and paste it in the space provided.

(Art Integration)



11 Types of Diseases

Rohan returned from school one afternoon and immediately ran to his mother. "Do you know why Rajat was not coming to school for the last ten days?", he asked her. "Vijaya Ma'am told us that Rajat is in hospital. He has got malaria. Some of us are planning to visit him this evening. Can I also go and see him, Mama?"

Rohan's mother gave him permission and it was decided that she would take all his



friends to the hospital. That evening on reaching the hospital, they saw Rajat looking pale and lying quietly on bed. On seeing his friends, Rajat cheered up a little. The nurse told the children to talk softly so as not to disturb the other patients. Soon the doctor came.

He was smiling, "Rajat I have some good news for you. Your blood reports have arrived. They are **negative**. The malaria is now under control. You will soon recover completely."

Rajat gave a big smile and the children clapped. It was good news indeed! But, the children wanted to know what malaria actually was. The doctor was a kind gentleman. He decided to spend sometime with the children to explain and enlighten them about this illness.



Put into Focus

- ▶ What causes malaria?
- ▶ Symptoms of malaria
- ▶ Methods by which malaria can be prevented
- ▶ Which are the other waterborne diseases?
- ▶ How can waterborne diseases can be controlled?



The doctor explained that malaria is a very serious disease. It is also the most common infectious disease in the world. In fact, malaria has affected humans for over 50,000 years.

“But how do people get infected?” the children wanted to know.

The doctor explained that malaria is caused when a person is bitten by a mosquito carrying a type of parasite called **plasmodium**. “Why is it called an infectious disease?” asked Rohan “How does it spread from one person to another?”

“Well, the doctor said, “If a female anopheles mosquito bites someone carrying the malaria parasite and then bites you, it will pass on the infection to you.”

After that, the malaria parasite passes through your blood into the liver where it grows and develops. Then, they attack the red blood cells and cause malaria.

At last the children understood that mosquitoes are responsible for spreading malaria.

Diagnosis of malaria

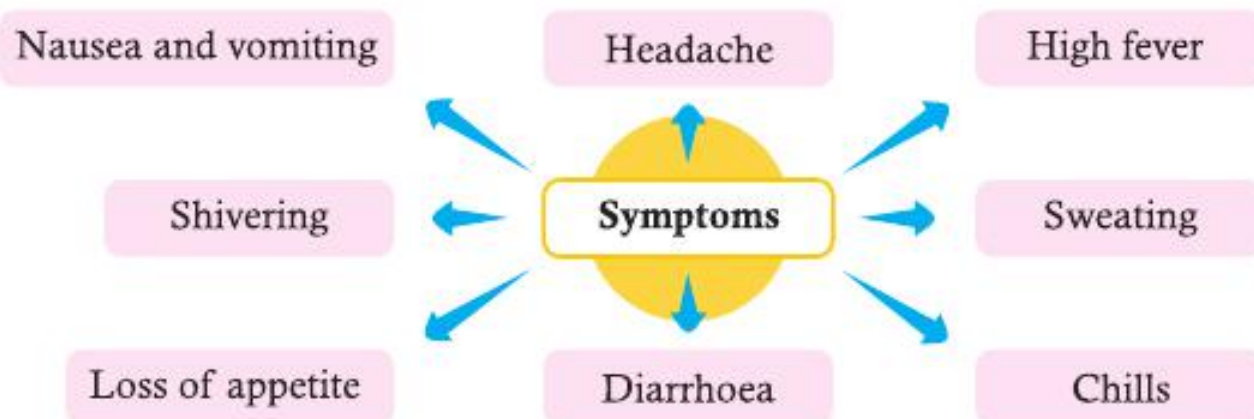
How do people know when they have malaria?

The doctor explained that there are some common features that help to diagnose malaria. The following chart shows the major symptoms that accompany the disease.



Know More

It was Sir Ronald Ross who proved in 1898 that malaria is transmitted through mosquitoes. He proved this while working in the presidency general hospital in Calcutta. He was awarded the Noble Prize in medicine in 1902 for his remarkable work on malaria.



When the above symptoms are present, the person is advised to get a blood test. If the test report is **positive**, you know the person has malaria.

Discuss with your teachers/parents and find out the similarities and differences between normal fever and malaria.

Cure

Once diagnosed, doctors prescribe medicines to cure malaria. Some of them are based on **quinine**.



Unfortunately, there is currently no approved vaccine available for malaria. But, scientists and doctors all over the world are working to find one.

Malaria can be cured in two days. Recovery can take weeks and sometimes months.

Prevention is better than cure

Since we know now that mosquitoes help in spreading malaria,

let us look at the following ways to protect ourselves from this diseases.



- ◆ Keep mosquitoes away by using mosquito repellent.
- ◆ Use mosquito net at night.
- ◆ Keep windows and doors shut early in the morning and evening.
- ◆ Wear long pants and long sleeves in the evening.
- ◆ Don't let water accumulate a keep for long in coolers, ditches, ponds etc.
- ◆ Sprinkle kerosene oil on stagnant water as it forms a thin film over the water bodies and prevents breeding of mosquito larvae.



Know More

The earliest treatment of malaria was the bark of the **cinchona tree**. It contains quinine out of which the medicine is made. It was first discovered in the jungles of South America. With the help of your teacher, identify South America on the Atlas.



Know More

25th April is the World Malaria Day. After biting a person the mosquito will rest on the walls of the house for sometime. Guppies are fish that eat the mosquito larvae in the water. They can save many people from getting malaria.

Water borne diseases



Mosquitoes lay their eggs in dirty water or in muddy places. Dirty or contaminated water is the breeding ground for various diseases causing bacteria, viruses and other small organism.

Water borne diseases are commonly transmitted through water supply contaminated by human, animal or chemical waste.

Water borne diseases

Diarrhoea

Cholera

Dysentery

Jaundice

Typhoid

Remember

Malaria occurs mainly between the months of June and November.



Know More

In developing countries, four-fifth of all illnesses are water borne. Diarrhoea is the most common waterborne disease and the main cause of childhood deaths.

So, find out where does the water you drink come from and whether, it has been made safe to drink.

Each year about three million people die prematurely from waterborne diseases around the world.

Often recreational areas like lakes, pools etc. too are responsible for spreading waterborne diseases, if they are not cleaned and disinfected regularly.

How can we check the spread of waterborne diseases?

Spreading of waterborne diseases can be checked by :

- ◆ Improving public sanitation facilities. Increase in the number of covered public toilets, to prevent the growth and spread of micro organisms.
- ◆ Prevent the discharge of any kind of waste in water bodies like rivers, lakes etc.
- ◆ Prevent personal hygiene.
- ◆ Provide clean drinking water that is free from any disease-causing germs. Use boiled or filtered water for drinking.

Infectious diseases caused by bacteria, viruses and small organisms are the most common and wide spread health risk associated with drinking water.



KEY WORDS

Parasite: Animal or plant that lives on another organism.

Boiling: Reach a temperature at which it bubbles and turns to vapour.

Larvae: A form taken by certain insects before metamorphosing into an adult.



OVERVIEW ON THE CHAPTER

- ▶ Malaria is spread by the female Anopheles mosquito.
- ▶ It is caused when a person is bitten by a mosquito, carrying a parasite called Plasmodium.
- ▶ Sir Ronald Ross proved that malaria is caused by mosquitoes. He received the Nobel Prize in Medicine for his discovery.
- ▶ Simple steps can help to prevent malaria.
- ▶ Consuming dirty water can lead to a number of dangerous and fatal diseases.

Assignment



COGNITIVE SKILLS



Multiple choice questions :

(Learning by Doing)

Tick (✓) the correct option.

- When a mosquito bites a human, the _____ enter the body.
a. parasite ☐ b. germs ☐ c. mosquito ☐ d. none of these ☐
- The parasites attack the _____.
a. red blood cells ☐ b. white blood cells ☐ c. plasma ☐ d. none of these ☐
- Malaria is due to the anopheles mosquito. The mosquito is _____.
a. male ☐ b. female ☐ c. animal ☐ d. none of these ☐



Fill in the blanks :

- The earliest known treatment of malaria was the bark of the _____ tree.
- _____ is a waterborne disease which mainly cause childhood deaths.
- Simple steps can help prevent _____.
- Consuming dirty water can lead to a number of dangerous and _____ diseases.



Answer the following questions :

- Name the mosquito which causes malaria. What are the symptoms of malaria?

- What is plasmodium? How does it cause malaria?

- How can malaria be prevented?

- What are waterborne diseases?

- How can waterborne diseases be prevented?

CRITICAL THINKING

Unjumble the following words to get the correct word :

(Self-Awareness)

- Mduimosalp : _____
- Hpeaonsle : _____
- Chincano : _____

4. Nujaidec : _____
5. Achordiar : _____

OBSERVING SKILLS

(Perceptual Thinking)

Below are given some symptoms. Some are indicators of malaria. Others are not.
Colour the symptoms of malaria with a red crayon.

Shivering

Pain in the eyes

Diarrhoea

Fainting

Vomiting

LINKING CHAIN

Find the correct answers in the puzzle below with the help of the given hints.

(G.K.)

N	O	B	T	N	P	H	E	E	L	O	T
C	H	O	L	E	R	A	Y	O	C	H	I
L	Z	I	A	N	O	P	H	E	L	E	S
X	N	L	W	A	T	E	R	N	E	V	X
Z	F	I	L	T	E	R	I	N	G	W	Y
O	T	N	B	A	K	Z	F	O	G	U	Z
P	L	G	U	P	P	I	E	S	S	U	Q
L	S	R	O	N	A	L	D	R	O	S	S

1. The most common way through which humans get diseases.
2. Water can be made safe by following one of these processes.
3. A disease caused due to dirty water.
4. Fish that eat mosquito larvae.
5. He proved that malaria is transmitted through mosquitoes.
6. The mosquito which is responsible for spreading malaria.
7. Mosquitoes lay these in dirty water.

12 Space

It was quiet night. Aditi was sitting with her father out in the balcony. Suddenly she asked, "Are there people living out on any other planet? Perhaps, on the moon."

Her father replied, "Well, as far as we know, the earth is the only planet in the solar system that has life."

"Why so?" asked Aditi.

Her father said, "It is because the earth has a very precious resource called water that supports life. Three-fourth of the earth is covered with water. That is why, it is called a **Watery Planet**. Besides, the earth is neither too far nor too close to the sun. It means that the temperature is just right for life to begin and flourish in the ample sunlight that we get.

"And yes, I know we also have oxygen," said Aditi "very good," said her father. "We have an **atmosphere** which contains important gases like oxygen, carbon

dioxide and nitrogen, along with water vapour."

This layer of air is vital for the survival of all living things. It also protects us from the harmful ultraviolet radiation of our nearest star, the sun.

The vast and limitless region beyond the earth's atmosphere is called **space**. The collection of information about the various heavenly bodies in outer space is called **space exploration**. We have always been fascinated by the moon, the sun and the stars. **The universe includes the earth and all the heavenly bodies in the space surrounding the earth.** The universe includes everything that exists in nature.

Solar system

The solar system consists of the sun, the eight planets and their satellites and other, heavenly bodies such as comets and asteroids. These eight planets including the earth



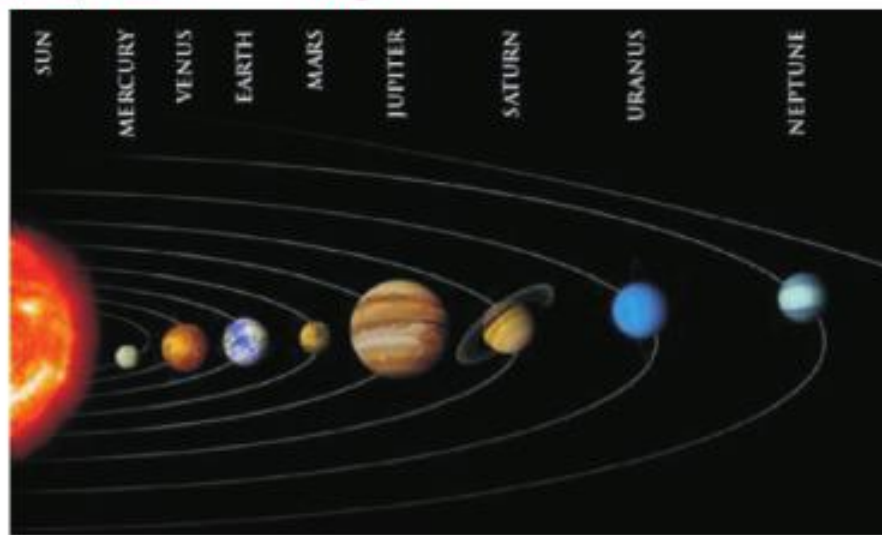
Put into Focus

- ▶ Solar system
- ▶ The Moon
- ▶ Artificial satellites



Earth

move around the sun. Each planet moves around the sun in its own path called **orbit**. The eight planets according to their distance from the sun are **Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus** and **Neptune**.



Solar system

The moon

The moon is the only natural satellite of the earth. It revolves around the earth in 27 days. It is nearest neighbour in space. It is 3,84,400 km away from the earth. It is the brightest object in the sky, next to the sun. It has no light of its own. It shines because it reflects the light of the sun.

The surface of the moon is rugged. It is made up of large craters (deep holes) and high mountains. The moon **has neither air nor water. So, no life exists on the moon.**



Moon

Artificial satellites

Man has launched a number of satellites into space. They move around the earth just like the moon. They are called artificial satellites.



Yuri Gagarin

1. Major Yuri Gagarin, a Russian, was the first man to go into space. Valentina Tereshkova, another Russian, was the first woman to go into space.



Valentina Tereshkova

2. The moon is our nearest neighbour in the sky. It has been studied more than any other thing in the sky. The moon is the first member of the solar system that has been visited by humans.

3. Squadron leader **Rakesh Sharma** became the first Indian to go into space. **Kalpana Chawla** became the first Indian woman to go into space in the U.S. space shuttle **Columbia**. **Ms. Sunita Williams** who is of Indian origin, has set a record for the longest stay in space by a woman. She left her space shuttle **Atlantis**, to go for a space walk! In other words, she walked in the space just outside her space shuttle.



Rakesh Sharma



Kalpana Chawla



Sunita Williams

4. **Space probes** : Astronomers use space probes (or space crafts) to study heavenly bodies. They land on a planet or revolve around it.

For example: The **Voyager Space Craft** has sent photographs of all the planets of our solar system from a very close range.



Know More

According to the international Astronomy union, the solar system consists of eight planets. Pluto, once considered to be a planet of our solar system, was dropped as it did not fit into the new definition of planets.

5. **Telescopes** : Some telescope like the Hubble **Space Telescope** are permanently stationed in outer space to observe various heavenly bodies.
6. **Space station** : It is a permanent space port where astronauts live and carry out scientific experiments in space. The **International Space Station** (ISS) is the only space station.
7. **Space shuttle** : It is a reusable spacecraft designed for sending astronauts into space. **Discovery** and **Columbia** are some well known space shuttles.



ISS



Discovery



Columbia

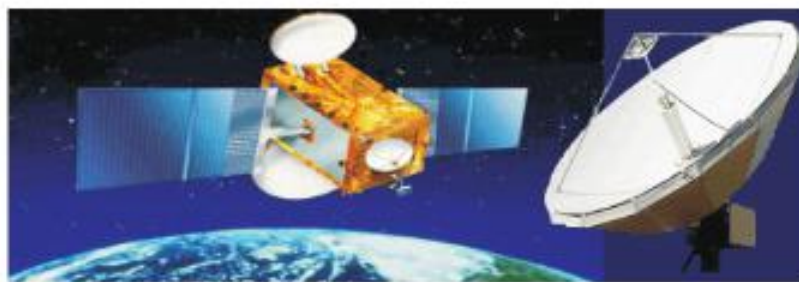
Uses of artificial satellites

Artificial satellites serve many useful purposes. Some of them are :

Communication satellites



INSAT-1



Edusat

They are used to transmit long distance telephone calls (STD and ISD) and television programmes around the world.

Weather satellites

They are used for weather forecasting. This information is used to warn people about floods, cyclones, storms and other weather related disturbances.

Space satellite

They are used to study heavenly bodies, stars, planets and moons.

Exploring the universe



If we want to discover more about something, we often try to get closer, to look for ourselves. **Astronomers** are scientists who spend their lives finding out about the stars and planets.

Astronomy is the science of studying space.

The stars and most of the heavenly bodies are too far away to be visited, so astronomers study the universe from a long distance. They do so with the help of telescopes.

A **telescope** is an instrument, that is used to study distant object and stars.

Thus, information is collected by the earth and space based **telescopes**. Astronomers usually work in places called **observatories** from where they observe the sky.



Know More

The universe is too big to measure in kilometres. Even, if you could travel at the speed of light, it would take at least 15 billion years to cross it as far as we know.



Know More

The first living creature in space was a Russian dog called Laika. She made a seven days journey in the space capsule, sputnik 2, in November 1957.



KEY WORDS

Space exploration: Space exploration is used in astronomy and space technology to explore outer space.

Crater: A large hollow in the ground.

Atlantis: A legendary island, beautiful and prosperous, which sank into the sea.



OVERVIEW ON THE CHAPTER

- ▶ The moon is the natural satellite of the earth. It revolves around the earth in 27.3 days.
- ▶ Artificial satellites are placed in definite orbits around the earth with the help of rockets.
- ▶ Rakesh Sharma, Kalpana Chawla and Sunita Williams are some famous astronauts of Indian origin.



Assignment



COGNITIVE SKILLS



Multiple choice questions :

(Learning by Doing)

Tick (✓) the correct option.

- The only planet where life exists _____.
 a. Venus ☐ b. Mars ☐ c. Earth ☐ d. Mercury ☐
- The farthest planet from the sun _____.
 a. Jupiter ☐ b. Neptune ☐ c. Uranus ☐ d. Mars ☐
- A record for the longest stay in space by a woman was set by _____.
 a. Velentina Tereshkova ☐ b. Sunita Williams ☐
 c. Kalpana Chawla ☐ d. None of these ☐

4. Our nearest neighbour in space is _____.

a. Sun

b. Mars

c. Saturn

d. Uranus



Fill in the blanks :

1. Man uses space probes to study _____.

2. The _____ has sent photographs of all the planets of our solar system.

3. _____ and _____ are some well known space shuttles.

4. Full form of ISS is _____.



Answer the following questions (short) :

1. What is universe?

2. What is meant by space exploration?

3. What is Sunita Williams' achievement?



Answer the following questions (long) :

1. Who were the first man and the first woman to go into space?

2. Name the first two men to walk on the moon.

3. What are artificial satellites? Name any one.

4. What are space shuttles? Name any two.

OBSERVING SKILLS

Circle the odd one in each group. Give the reason for your answer.

(Perceptual Thinking)

1. Rakesh Sharma

Kalpana Chawla

Sunita Williams

Yuri Gagarin

2. Space probes

Space station

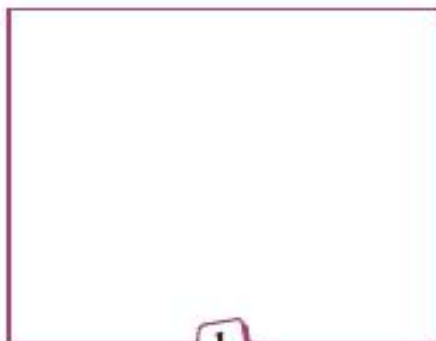
sun

Space shuttle

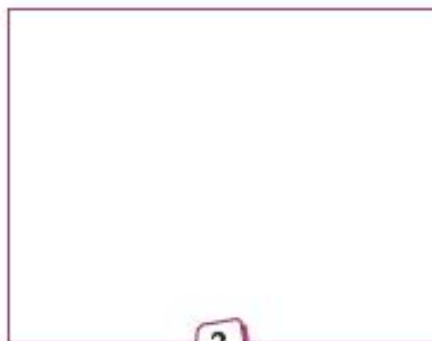
Collect pictures of various planets, stars and the moon.

Paste them in the space provided.

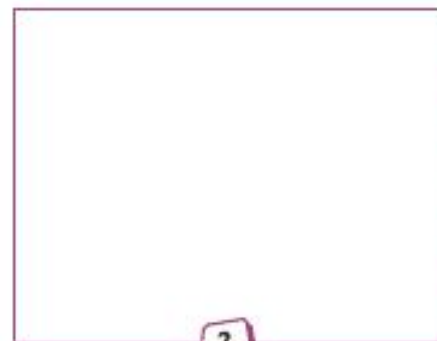
Write five sentences about each.



1.



2.



3.

1. Planet

2. Stars

3. Moon

13 Every Drop Counts

All of us need fresh water for drinking and cooking purposes. Water present on the earth's surface is not fit for drinking as it contains many impurities. The source of water supply in a city is usually a nearby lake or a river. The river and lake water usually contains various salts as impurities. Salt is removed from impure water to make it drinkable. For this, water purification plants are set in every city. Water from different water-bodies like rivers, ponds etc is brought to the purification plant for filtration and chemical purification. Purified water is supplied to the houses.

Clean drinking water

Water needs to be purified to make it drinkable water, that is suitable for drinking, is called potable water.

Water from various sources is purified. We follow the following steps for purification of water.

Sedimentation

In this step, water is allowed to stand still so that the heavy particles of impurities present in it can be settle down. This process is called sedimentation.



Water purification plant

Filtration

Sedimentation is followed by filtration. In this step, water is passed through chemicals, sand and gravel, that act as filters to remove the finer particles of impurities. This process is called filtration.

Chlorination : After filtration, a small amount of chlorine is added to disinfect the water and kill the germs present in it. This process is known as chlorination. The potable water is then stored in water

tanks. It is supplied to houses through pipes and taps.



Put into Focus

- ▶ Clean drinking water
- ▶ Water pollution
- ▶ Effects of water pollution
- ▶ Water is scarce



Water drop

Water pollution

Fouling of water by human activities is called **water pollution**. Water pollution makes drinking water dirty and harmful for us. The water of the river is made dirty by various human activities- bathing, washing clothes and dumping the wastes. Dirty water is said to be polluted.

Causes of water pollution

The main causes of water pollution are domestic and industrial waste thrown into water sources.

Water is polluted by any of the following :

- ♦ harmful pesticides used by farmers.
- ♦ chemicals from soaps and detergents used in washing.
- ♦ harmful chemicals from industries.
- ♦ animal's waste.

Effects of water pollution

Water pollution harms us in many ways.

Polluted water contains germs that cause various diseases **diarrhoea**, **typhoid**, **cholera**, **dysentery**, **hepatitis** and **jaundice**. Apart from these diseases, it can also cause skin diseases.

Water pollution has harmful effects not only on humans but also on aquatic life. A large number of fish and other aquatic animals die due to polluted water. Oil and toxic waste from industries, when thrown into water bodies, can cause the death of aquatic plants and animals.

Water pollution reduces the oxygen present in the water. The reduction in oxygen is harmful for the living beings in water.

Water is scarce

Some parts of the world always have shortage of water. Some people try to solve this problem by digging deep wells. Drawing excess water from underground brings down the water level. So, to reduce water scarcity is a big problem.

Rain water, the main source of water, can be conserved through **rainwater harvesting**.

Rainwater is collected in tanks and passed through filters to remove impurities.



Know More

Salt is obtained from sea water. To get the salt from the water, a low wall is built around the shallow section of the sea where sea water is trapped. The water evaporates under the hot sun, leaving behind layer of salt in India, you will find such salt farms in the coastal plains of Gujarat.



Polluted water

Then it is stored and used for drinking and other activities.

Harvested rainwater recharges groundwater. This technique helps to reduce scarcity of water.



KEY WORDS

Sedimentation: Matter that settles to the bottom of a liquid.

Filtration: Filtering liquid.

Chlorination: Treat water with chlorine.



OVERVIEW ON THE CHAPTER

- ▶ Water present on the earth's surface is not fit for drinking.
- ▶ The river and lake water contains various salts as impurities.
- ▶ The main causes of water pollution are domestic and industrial waste thrown into water bodies.
- ▶ Water pollution has harmful effects on all living beings.
- ▶ Water suitable for drinking is called potable water.
- ▶ Every city has a water purification plant to obtain potable water.



COGNITIVE SKILLS



Multiple choice questions :

(Learning by Doing)

Tick (✓) the correct option.

1. _____ present on the earth's surface is not fit for drinking.
a. water ☐ b. land ☐ c. air ☐ d. salt ☐
2. The _____ and _____ water contains various salts as impurities.
a. river ☐ b. lake ☐ c. pond ☐ d. ocean ☐
3. When heavy particles of impurities present in water can be settle down is known as _____.
a. sedimentation ☐ b. filtration ☐
c. chlorination ☐ d. none of these ☐



Fill in the blanks :

1. _____ is removed from impure water to make it drinkable.

2. Polluted water contains _____ that cause various _____ in human beings.
3. Water pollution reduce the _____ present in the water.
4. Water suitable for drinking is called _____ water.
5. Scarcity of water can be reduced by _____ rain water.



Define the following terms :

- | | |
|------------------|--------------------------|
| 1. Potable water | 2. Water pollution |
| 3. Chlorination | 4. Rain water harvesting |



Answer the following questions :

1. What is water pollution?

2. How do water bodies get polluted?

3. What kinds of diseases caused by polluted water?

4. Write the steps to purify water.

5. What is potable water?

CRITICAL THINKING

(Self-Awareness)

List the ways in which the water sources near your area become dirty. Mention, how can you reduce it? Write in around 150 words.

14 The Trees



Trees

Trees and plants not only make our surroundings green but also purify air. In the food making process, plants take in carbon dioxide and give out oxygen. Many birds and animals build their homes in

trees. Trees also provide us fruits and medicines.

There are many plants that we grow in our garden such as **rose**, **pansy**, **jasmine**, **marigold** and **money** plant. Ashoka and rubber trees are planted for decoration. They also add greenery to the house.

Trees in our surroundings

Neem, **peepal**, **jamun**, **magnolia**, **gulmohar**, **laburnum**, etc. are some commonly seen trees in our surroundings.



Neem Tree



Gulmohar tree



Magnolia tree



Jamun tree

1. Neem is a good air purifier. Leaves, bark and sap of neem tree have medicinal value. Many toothpastes and powders are prepared by using neem leaves.
2. Gulmohar tree bears red flowers and give shade.



Put into Focus

- ▶ Trees in our surroundings
- ▶ Wild plants
- ▶ Plant care



Plants



Know More!

The dwarf varieties of trees grown in trays or pots, used for the purpose of decoration in houses, called bonsai.

3. Magnolia tree bears white or pink flowers with a sweet fragrance. Beautiful fragrant flowers of magnolia are used for worship.
4. Jamun tree is also a commonly seen tree. Its berry type fruits called Indian blackberries(jamun) are eatable and are liked by many people. They are also used in making medicines.
5. At some places, we can see huge shady trees with hanging roots. These trees are banyan trees. Their branches spread out widely.



Banyan Tree



Peepal Tree

6. Peepal tree is also seen commonly. It is worshipped by Hindu people. This tree grows wherever its seeds fall.

7. The **babool** or **kikar** tree is often seen in the countryside. It can grow easily even in dry weather. Its hard wood is used by people to make ploughs. Cattle, sheep and goats eat its pods.



Babool Tree

Wild plants

The plants, that grow themselves and nobody seems to take care of them except nature, are called **wild plants**. Plants such as **weeds**, **water hyacinth** and **daisy** grow wild.

Some of the wild plants are so dangerous that they cause skin allergy even when touched a little, while some are useful. Some of them increase fertility of soil. Besides, most of them provide food and shelter for birds and insects.

Plant care

The plants, that we grow in or around our house, are taken care by us or by the gardener employed by us. Similarly, plants grown in schools or other official buildings are looked after by gardener. Trees along the streets or roadsides are planted by the municipal authorities. Each year, our government plant thousands of saplings.



Plant sapling

Saplings are protected from cattle by encasing them in metallic tree guards.

Municipal authorities also take care for regular watering and manuring of these plants. They also make arrangements for saving those trees from pests and insects.

Trees that grow in forest, are protected by the employees of the forest department.



Tribal people

Tribals that live in forests also help in protecting forest trees though they totally depend on plant products for their food, shelter, fuel, fodder, clothes and medicines. They take care that trees, plants are not over exploited.

They use plant products of a certain area for a short period and then they leave the place and shift to some another place. In this way, that area maintains its natural balance. Tribals also take care to nurture the forests. They live in balance and harmony with the forests. Tribals are the real environmentalists. All of us should make efforts to stop cutting of trees and grow more and more trees.



Watering plants



KEY WORDS

Sapling: A young tree.

Manure: Animal dung used as fertilizer.

Tribal: A social group in a traditional society consisting of linked families.



OVERVIEW ON THE CHAPTER

- ▶ Trees and plants not only make our surroundings green but also purify air.
- ▶ Trees provide us many useful things like fruits, medicines etc.
- ▶ Animals and birds build shelter in the trees.
- ▶ The plants that grow themselves and nobody seems to take care of them except nature, are called wild plants.
- ▶ Tribals live in balance and harmony with the forests.
- ▶ All of us should make efforts to stop cutting of trees and grow more and more trees.

Assignment

COGNITIVE SKILLS



Multiple choice questions :

(Learning by Doing)

Tick (✓) the correct option.

- _____ and _____ trees are planted for decoration.
a. Ashoka ☐ b. Rubber ☐
c. only (a) and (b) ☐ d. none of these ☐
- _____ tree is worshipped by Hindu people.
a. Jamun ☐ b. Banyan ☐ c. Peepal ☐ d. Gulmohar ☐
- _____ tree grows easily even in dry weather.
a. Peepal ☐ b. Babool ☐ c. Jamun ☐ d. Gulmohar ☐



Fill in the blanks :

- _____ and _____ not only make our surroundings green but also purify air.
- Gulmohar tree bears _____ and give _____.
- Trees along the streets or roadsides are planted by the _____.
- Trees, that grow in forests, are protected by the employees of the _____.



Write briefly about each of the following :

- Jamun tree : _____
- Neem tree : _____
- Banyan tree : _____
- Peepal tree : _____
- Babool tree : _____



Answer the following questions :

- How are plants useful to us?
- Define wild plants.

ACTIVITY & Critical Thinking

(Art Integration & Self Awareness)

Sketch some of the trees that you see in your neighbourhood. Write their names below them. Also mention whether they produce edible fruits or not.

15

Work, Force and Energy

Force

When you get up from your chair observe your movements carefully. First you push against the floor and then pull yourself up. These push and pull movements cause things to move. When we push an object, it moves away from us. But when we pull an object, it comes closer to us.

This push or pull applied on a body is called **force**. All movements are the result of some kind of force. There are many forces acting around and inside us. Example: force inside our body makes our heart beat.

Effects of force

It can move a resting object.



Pressing the ball

It can change the direction of a moving object.



Swing

It can also change the shape of an object.



Pushing the wheelchair



Playing badminton



Put into Focus

- ▶ Force
- ▶ Effects of force
- ▶ Kinds of forces
- ▶ Work
- ▶ Energy
- ▶ Sources of energy
- ▶ Forms of energy

It can stop a moving object.



Catching a ball



Catching a ball

Kinds of forces

Force can be of different kinds.

Muscular force : When we push a swing or catch a moving ball, we use the muscles of our arm. This is called **muscular force**.



Iron pin attracted by magnet

Magnetic force : Bring a magnet near a few iron pins. The magnet pulls iron pins to it. This force applied by the magnet is called **magnetic force**.



Catapult

Elastic force : This is applied when we use a catapult to throw a stone. When we stretch the rubber strip of the catapult, it applies force on the stone. This force is called **elastic force**.

Mechanical force : The force applied with the help of tools is called **mechanical force**. E.g., a pair of scissors to cut a piece of paper or a bottle opener to open a bottle.

Gravitational force : The force by which the earth pulls the objects towards it, is called **gravitational force**. e.g. balls are falling down.



Balls are falling down

Work

When force is used on an object, there may be a change. The object may start moving, stop moving or change direction. When any of these things happens, work is said to be done. **Work is said to be done only when a force (pull or push) moves a body through a distance.**

For example: When a child pushes a chair and the chair moves, work is said to be done, but when he/she pushes a cupboard and the cupboard does not move, no work is being done.



Movement



No Movement

Energy

Energy is the ability or capacity of a body to do work. All living things use energy to work. Energy can be changed from one form to another.

Example– A call bell turns electrical energy into sound energy.

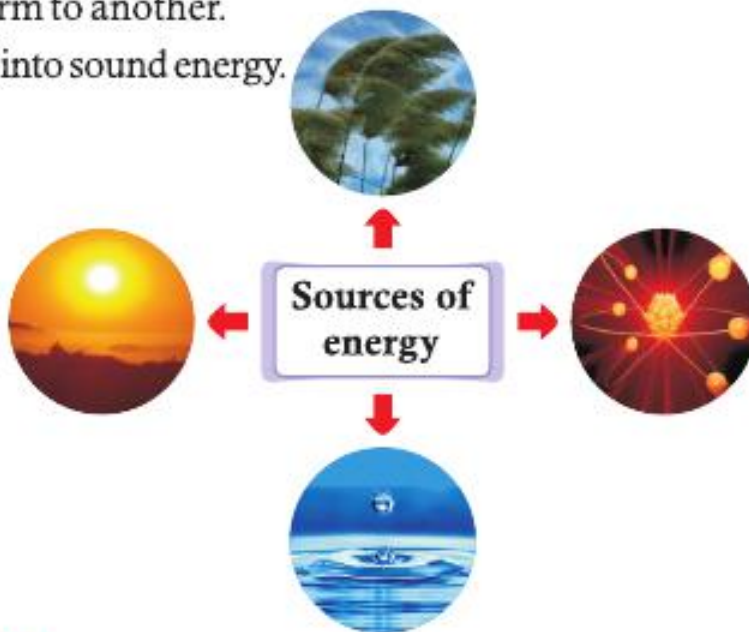
Sources of energy

Sun : The heat of the sun is the main source of energy on the earth.

Wind : Wind energy is produced by movement of air.

Water : Water energy is produced by movement of water.

Atom : This is the source of energy about which you will learn in your higher class.



Muscular energy

Forms of energy

Muscular energy

When we lift, pull or push an object, we use energy stored in our muscles. The energy stored in the muscles of human beings or animals is called **muscular energy**.

Wind energy

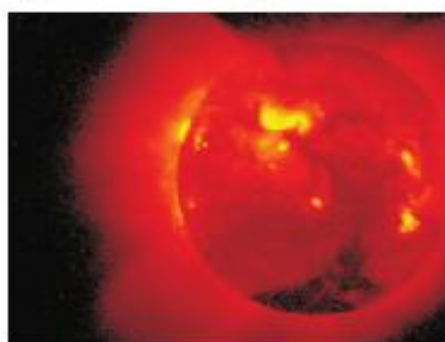
Moving air is called wind. Wind possesses energy called wind energy. Wind energy can be used by rotating the blades of a wind mill. This energy is used to run a pump to draw underground water and generate electricity.



Wind energy

Water energy

Flowing water has energy in it. Falling water from dams is used to rotate turbines to generate electricity. Electricity produced by the energy of flowing water is called hydroelectricity.



Heat energy

Heat energy

Energy present in a hot body is called heat energy. It is obtained by burning fuels like wood, coal and petrol.

The heat produced by burning petrol in a car-engine provides energy needed to run the car.

Electrical energy

The energy that is obtained from electricity is **electrical energy**. We use electrical energy to light electric bulbs and run different electrical appliances like geysers, irons, ovens and refrigerators.

Sound energy

Sound is also a form of energy.

Sound as a form of energy which is experienced when sound from an alarm clock or call bell travel to us.



Sound energy



Solar cooker



Solar heater

Solar energy

The energy that we get from the sun is solar energy. These days, solar energy is used in cookers and solar heaters for cooking and heating.



KEY WORDS

Force: Physical strength or energy.

Magnetic: Having the property of a magnet.

Elastic: Able to return to normal size or shape after being stretched or squeezed.



OVERVIEW ON THE CHAPTER

- ▶ There are five kinds of forces namely muscular force, magnetic force, elastic force, mechanical force and gravitational force.
- ▶ Sources of energy include wind, water, sun and atom.
- ▶ Forms of energy include muscular energy, wind energy, water energy, heat energy, electrical energy, sound energy and solar energy.

Assignment



COGNITIVE SKILLS



Multiple choice questions :

(Learning by Doing)

Tick (✓) the correct option.

1. An apple falling from a tree is due to _____ force.
a. gravitational ☐ b. magnetic ☐ c. elastic ☐ d. none of these ☐
2. The force applied with the help of tools is called _____ force.
a. muscular ☐ b. mechanical ☐ c. elastic ☐ d. magnetic ☐
3. We get solar energy from the _____.
a. cloud ☐ b. moon ☐ c. sun ☐ d. stars ☐
4. Refrigerators work on _____ energy.
a. heat ☐ b. electrical ☐ c. motor ☐ d. wind ☐



Fill in the blanks :

1. Force can change the _____ of a moving object.
2. The force applied by a magnet is called _____.
3. _____ is the ability or capacity of a body to work.
4. Wind energy can rotate the blades of a _____.



Name the types of energy used in each of these pictures.





Answer the following question :

1. Define force.

2. What is magnetic force?

3. When is the work said to be done ?

4. Name four sources of energy and four forms of energy.

5. Name the kind of force used in each of the following examples.

a. a person drawing water from a well.

b. magnet pulling iron pins towards it.

c. effort applied to open a bottle.

d. an apple falling from a tree.

OBSERVING SKILLS

Circle odd one in each group.

1. geyser

oven

bicycle

refrigerator

2. water

coal

goat

petroleum

(Perceptual Thinking)

ACTIVITY & Critical Thinking

(Art Integration & Self Awareness)

Collect different types of toys. Study their working. Write about their source of energy.

16

Solids, Liquids and Gases



Put into Focus

- Matter on the earth
- Different forms : solid, liquid and gas
- Knowledge of interchange between different stages of matter

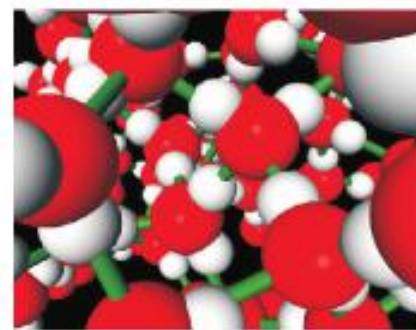
Friends, we have read in previous classes that anything that has weight and occupies space is called **matter**. We also know that the matter on the earth is found in three different forms—solid, liquid and gas. Now, could you recall what is matter made up of? Yes! the 'matter' is made up of very small particles, called molecules.

Let us go ahead and learn more about it.

Molecules

Molecules are very small particles that make up the different substances found on the earth. Molecules are so small that we cannot see them with our naked eyes. We have to use a

microscope to see molecules in a substance. Large number of molecules



Molecules of a Substance

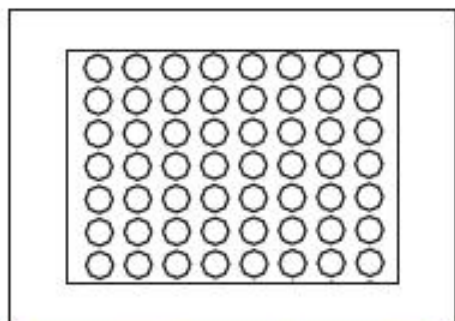
join each other to make a substance. The molecules have some spaces among them, known as **intermolecular spaces**.

The molecules of a substance are held together with a force attracting each other. This attraction or force among molecules differs in different substances. For example, molecules of solids have greater attraction, so they pull each other with a strong pull.

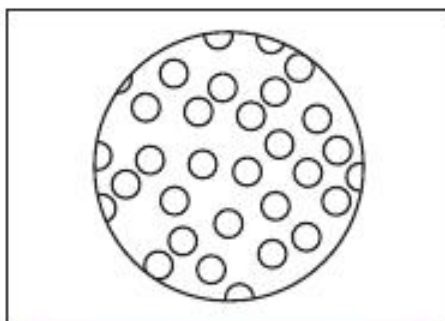
**Know More**

The ancient Indian seer, Kanada was the first to discover that all substances on the earth are made up of small or tiny things. He called these small or tiny things as Anu.

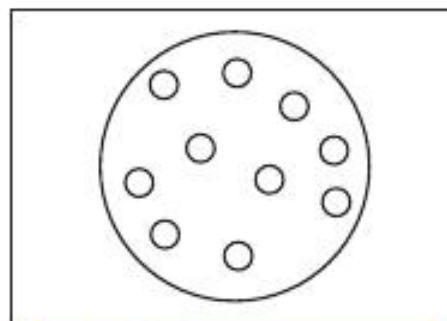
This keeps them tightly packed and the substance remains solid. The molecules of a liquid have less strong attraction or force among them. So, they stay a bit having more intermolecular spaces among them.



Molecules of solids stay tightly packed with negligible intermolecular space.



Molecules of liquid are a bit loose because they have lesser attraction among themselves and more intermolecular spaces.



Molecules of gases have the weakest attraction among them and stay quite far from each other. So the intermolecular spaces among them are much more than in any other state.

Molecules of gases have the weakest force of attraction among themselves, so they stay far apart from each other having wide spaces in between. Since the molecules of a gas have enough spaces in between, they can have more movement than the molecules of solids or those of liquid. It is for this simple reason that gases move at a faster rate even without slopes, while liquids need slopes to move (flow).

Solubility of water

As you have studied in previous class, the ability to dissolve is called solubility. Liquids enjoy a great deal of solubility while gases and solids do not. Water is called the **universal solvent** as it can dissolve a large number of substances.

Do you know how a liquid, say water, dissolves another substance into itself? To know just read on.

We know that every substance is made of molecules. Liquids also have molecules which have small intermolecular space in-between. When we dissolve a substance (solute) into a liquid (solvent), its molecules fit into the intermolecular spaces of the solvent. Thus, both the solute and the solvent mix so evenly that each of them ceases to appear as a separate substance.



Take a glass jar and fill it with marbles. Now can you see the spaces in between the marbles? These are some as the intermolecular spaces in a substance. Now pour some sand into the jar, and see where it rests finally. It settles in the spaces among the marbles. This is the same as the molecules of a solute settle in the intermolecular spaces of solvent.



Solids in liquids

Most of the solids like salt, sugar, etc. are soluble in water. The molecules of these substances settle easily in-between the molecules of water. Some other solids like sand, chalk, etc. are insoluble in water. Some solids like wheat, flour, etc. are partially soluble in water.

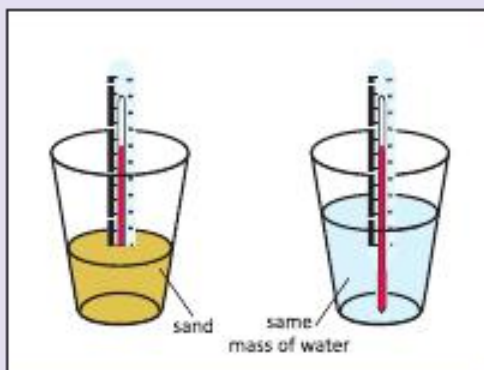


Liquids in water

Many liquids like glycerine, alcohol, etc. are soluble in water.



Take a glass full of water and some glycerine. Add a teaspoonful of glycerine to the water and stir it well. Observe the water carefully. What do you see? The glycerine mixes with in the water.



Some other liquids like mustard oil, petrol, kerosene, etc. are insoluble in water. Try to mix a teaspoon full of mustard oil in a glass of water and see if it dissolves there in. You will see that it does not dissolve.

Miscibility

The quality of liquids dissolving in liquids is called **miscibility**. If a liquid dissolves completely in another, it is called **miscible**. If a liquid does not dissolve in the other, it is called **immiscible**. For example, alcohol is miscible in water and petrol is immiscible in it.

Gases in water

Some gases like carbon dioxide, nitrogen, etc. dissolve in water. Aquatic plants and animals stay alive on oxygen that is dissolved in water. Cold drinks are a fine example of gases being dissolved in water. Take a bottle of cold drink and open it. Shake the bottle for a while. What do you observe? You see many air bubbles coming out of the cold drink bottle. From where do these bubbles come? The dissolved gases in the cold drink (water) come out as bubbles.

Thus, we see that gases can be dissolved in water.

Interchange of the states of matter

We have studied in previous classes that the 'matter' can change its states. Solids change into liquid on heating, and if continued to be heated, they finally turn into gases. Conversely, gases turn into liquids on cooling, and if cooled further, they become solid ultimately.

Thus, the main thing which decides the interchange of states of matter is temperature. Every substance changes its states at fixed temperature.

Boiling point

The temperature at which a liquid starts boiling is called 'boiling point' of that liquid. Water starts boiling at 100°C , so its boiling point is 100°C .

Melting point

The temperature at which a solid starts converting into a liquid, is called its **melting point**.

Freezing point

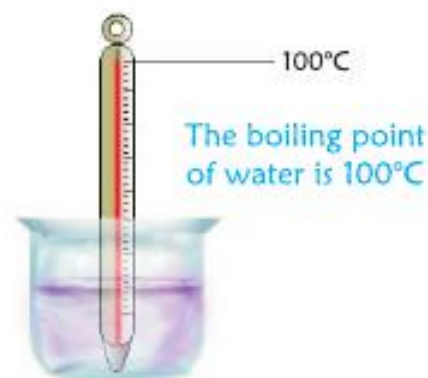


Know More

Metals need very high temperature to melt, high as much as 1000°C .



Aquatic Animals



The temperature at which a liquid freezes, is called its freezing point.



Melting

Water freezes at 0°C so, its freezing point is 0°C .

Let us see some common terms associated with the interchange of states of matter.

- When a solid converts into a liquid, the process is known as **melting**. Ice melts to form water through melting.
- When a liquid converts into a gas, the process is known as **evaporation**. It is for this reason that the process is the one in which 'vapours' are formed. The formation of water vapour or steam takes place through this process.
- When a gas turns into a liquid, the process is called **condensation**. The clouds turn into rain drops through this process.
- The conversion of liquid into a solid is called **solidification**. Formation of ice from water takes place through this process. It is also known as freezing.



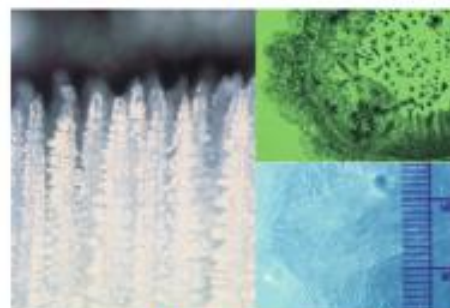
Water freezes at 0°C



Evaporation



Condensation



Solidification

Physical and chemical changes

A change which is limited to only external form of a thing is called **physical change**. A physical is a temporary change and a thing changed physically can turn back to its original form. For example, conversion of water into ice is a **physical change**. In a chemical change, even the molecular structure of a thing is changed. A thing changed chemically cannot turn back to its original form. For example, burning of a candle is a chemical change.



Physical Change

Chemical Change



KEY WORDS

Matter: Anything that has weight and occupies space is called matter.

Molecules: Molecules are very small particles that make the substances.

Intermolecular spaces: The molecules have some spaces among them, known as intermolecular spaces.

Miscibility: The quantity of liquids dissolving in liquids called miscibility.



OVERVIEW ON THE CHAPTER

- ▶ Matter consists of molecules.
- ▶ Molecules are microscopic and are held together by a force of attraction.
- ▶ The spaces among the molecules are called intermolecular spaces.
- ▶ Water dissolves most of substances, so it is called universal solvent.
- ▶ The temperature at which a liquid starts boiling is called its boiling point.
- ▶ The temperature at which a solid melts into a liquid is called its melting point.
- ▶ The temperature at which a liquid freezes into solid is called its freezing point.
- ▶ The temporary change in external form of a thing is called a physical change.



Assignment



COGNITIVE SKILLS



Multiple choice questions :

(Learning by Doing)

Tick the (✓) correct option.

1. Which of the following has weak force of attraction among the molecules?
a. solids ☐ b. liquids ☐ c. gases ☐ d. none of these ☐
2. Which of the following is soluble in water?
a. sugar ☐ b. chalk ☐ c. sand ☐ d. oil ☐
3. Which of the following is partly soluble in water?
a. salt ☐ b. wheat flour ☐ c. oxygen ☐ d. ghee ☐
4. The boiling point of water is :
a. 0°C ☐ b. 56°C ☐ c. 100°C ☐ d. 150°C ☐

5. What is the freezing point of water ?

a. 0°C

b. 56°C

c. 80°C

d. 100°C



Fill in the blanks :

solubility

solids

soluble

molecules

liquids

1. Molecules of _____ stay tightly packed.
2. Matter is made up of very small particles called _____.
3. The ability to dissolve is called _____.
4. Molecules of _____ are a bit loose as they have weaker force of attraction.
5. Most of the solids like sugar, salt, etc. are _____ in water.



Write T for true and F for false statements :

1. Liquids that mix into other liquids are called miscible liquids.
2. Solids melt into liquids on cooling.
3. Water turns into steam through evaporation.
4. Liquids turn into solids through condensation.
5. Coconut oil turns solid on cooling.

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Match the column A with the column B :

A

1. Liquid
2. Liquid freezes
3. Solid starts converting into a liquid

B

- a. freezing point
- b. evaporation
- c. melting point



Answer the following questions :

1. What is intermolecular space ?

2. What holds the molecules together in a substance ?

3. What is a miscibility change ?

4. What is physical change?

5. What is a chemical change?

PROJECT WORK

(Learning by Experiment)

Take some coconut oil and keep it in the refrigerator for 8 - 10 hours. Observe what happens to it thereafter.

17

Tides, Shadows and Eclipses



Put into Focus

- ▶ The tendency or quality which pulls everything towards earth's centre is called gravitational force.
- ▶ We should know about the formation of tides and shadows
- ▶ A substance through which you can see a thing clearly, is called transparent substance.
- ▶ An eclipse takes place only when the sun, the moon and the earth fall in a straight line.

Friends, we have read a lot about the heavenly bodies present in the space. We also came to know of several processes that occur due to the motions or positions of the heavenly bodies. The formation of day and night, change of seasons, etc. are some occurrences due to different positions of the sun, the moon and the earth. Let us learn about how each of these occur.

Tides

We know that our earth force of its own through which it pulls everything towards its centre. This tendency or quality is called **gravity** or **gravitational force**. Like the moon also has its own gravitational force. Since the moon is close enough to our earth, it pulls the water on the earth's surface (seas and oceans) towards its centre. This risen

part of water in seas and oceans is

called the **high tide**. This also happens on the opposite side of the earth. So there are two high tides at the opposite sides of the earth at a time.



The moon and the tides



Know More

Like the earth and the moon, the sun also has its gravitational force, but it does not affect things on the earth simply because it is very far from it.

There lies a low tide between the two high tides.

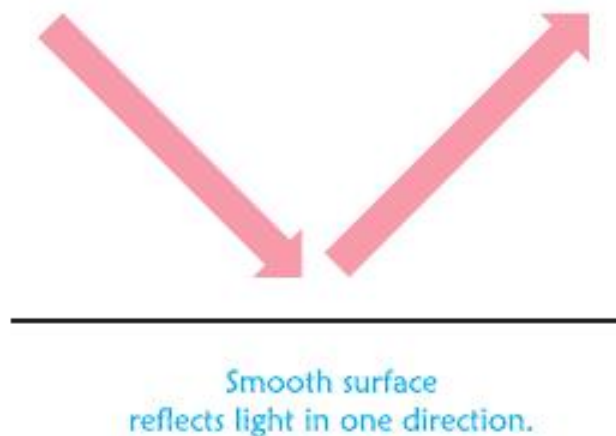
Reflection of light

We know that light moves in a straight path. When a light ray strikes a surface, it is reflected in opposite direction. A light ray falling on a flat surface is reflected at a right angle, while that falling on an uneven surface, is reflected in several directions.

Transparent and opaque substances

A substance through which you can see a thing clearly, is called **transparent** substance. On the other hand, the substance, through which light rays cannot pass, is called an **opaque** substance.

A substance that lets the light rays pass through it partially is called **translucent**.



Bumpy and uneven surface reflects light in all directions.



When light fails to pass an opaque substance, a dark patch forms on the other side of the body.

Formation of shadow

When light falling on opaque substance fails to pass it, a dark patch forms on the opposite side of the substance. This is called the **shadow**. It can be formed only in light as light rays are the originators of a shadow. This is why, we see shadows only during day time and in proper light during night. No shadows can form in complete darkness.

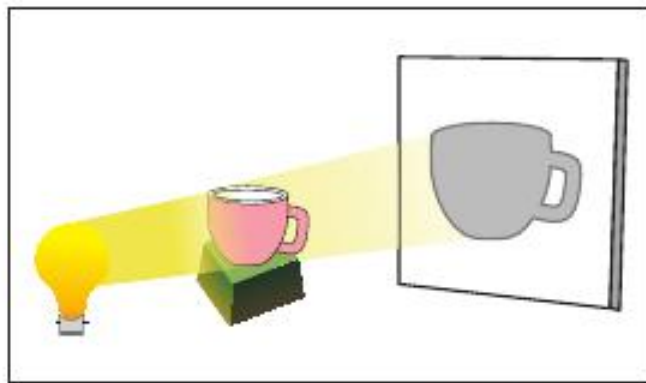
Shadows keep changing their sizes depends on the angle of light falling on a body. If light

falls on a body from sides, the shadow will be longer. If the source of light is right above the object, the shadow is longer in the morning and evening, and shortest in noon.

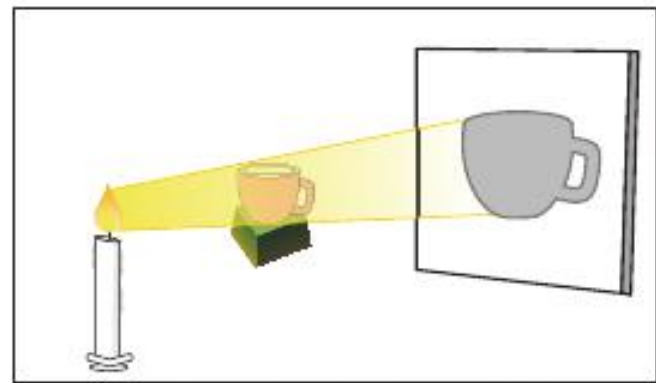


The size of shadows depends on angle of light

Shadows formed in a bright light are darker than those formed in dim light.

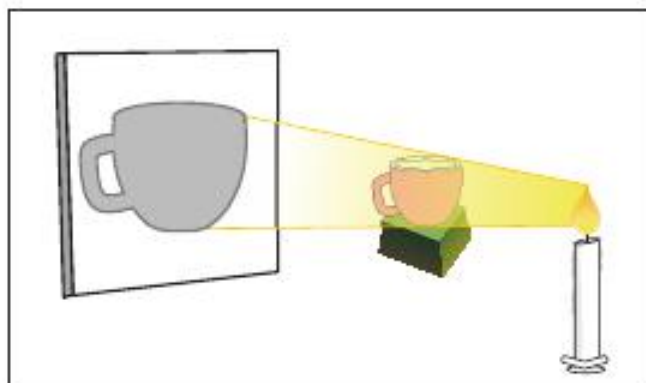


Shadow formed in a bright light

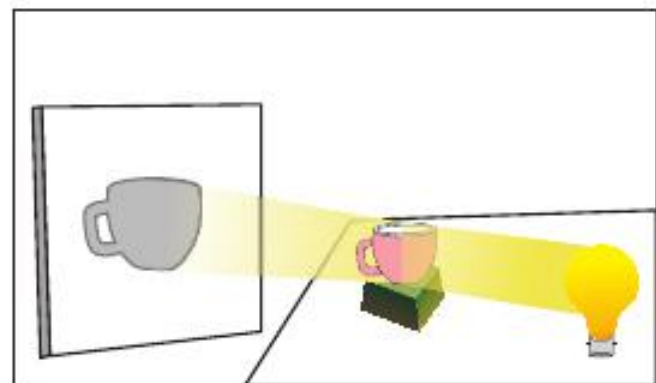


Shadow formed in a faint light

The size of a shadow depends on the size of the object.

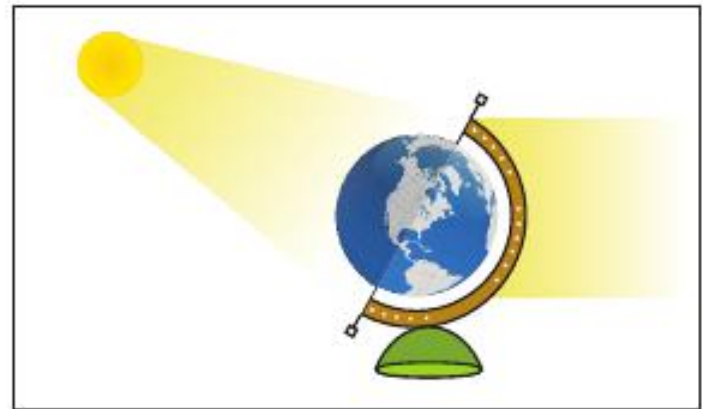
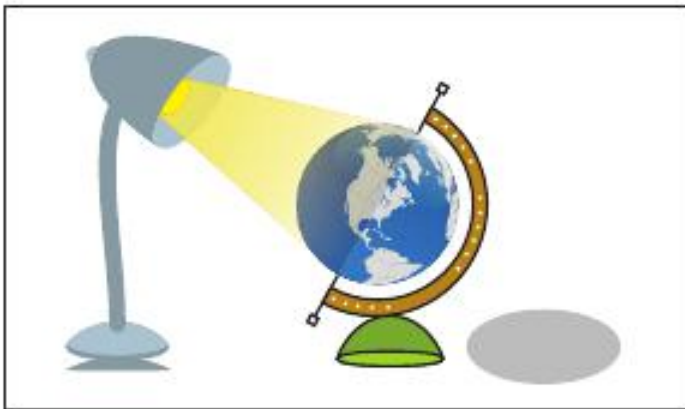


Shadow of a large object is large



Shadow of a small object is small

The size of a shadow also depends on distance between the source of light and the object. If the source of light is near the object, the shadow formed is small and if the source of light is far away from the object, the shadow formed is large.

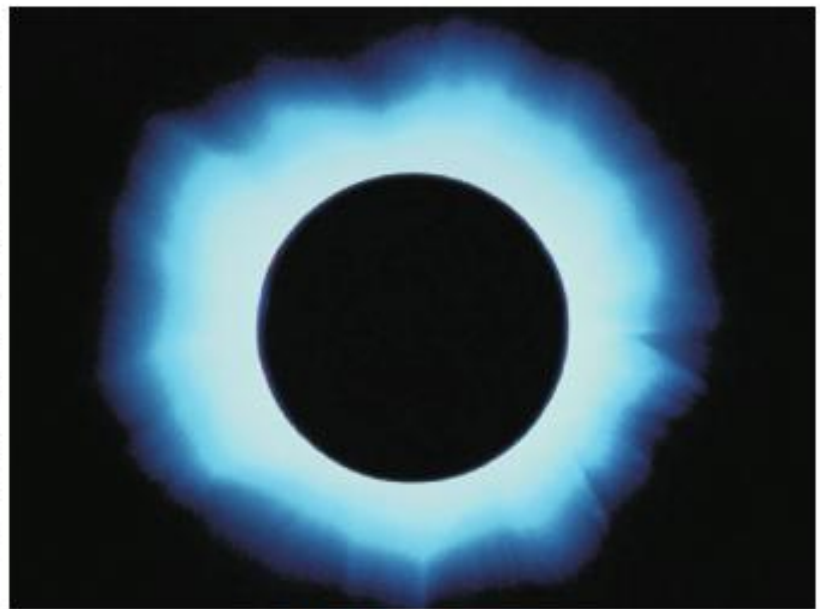


Eclipses

An eclipse takes place only when the sun, the moon and the earth fall in a straight line.

The Solar Eclipse

The earth revolves around the sun, and the moon revolves around the earth. During these motions, when the moon comes between the sun and the earth, and they all fall in a straight line, the shadow of the moon falls on the earth. This causes darkness at the affected part of the earth even during a bright sunny day. The people living in this part of the earth do not get the light of the sun in such a situation, or are able to see only a small part of it. If the sun goes out of sight completely, it is called total **solar eclipse**, and if only a part of the sun is visible, it is called a **partial solar eclipse**.



The Solar Eclipse

A solar eclipse occurs on a new moon day. It does not take place on every new moon day. By simple on every new moon day, the earth, the sun and the moon cannot be in a straight line.

As soon as the moon moves away from the face of the sun, the shadowed part of the earth, starts receiving the sunlight, and the eclipse is said to be over.

The Lunar Eclipse

When the earth comes in-between the sun and the moon it obstructs the sunlight from reaching the moon. Due to this, a part of or the whole of the moon falls in darkness. Thus, it goes invisible for the people on the earth. This is called the **lunar eclipse**. If only a part of the moon is visible, the eclipse is called the **partial lunar eclipse**.



The Lunar Eclipse



Know More

We should not try to see the solar eclipse with our eyes directly as it is harmful to our eyes. It is, thus better that we use some indirect way to see such an eclipse. Now-a-days eclipses are shown on television in completed details.

A lunar eclipse takes place on a full moon night. Lunar eclipse does not take place on every full moon night for the reason that on every moon night, the sun, the moon and the earth can not be full in a straight line.

Artificial satellites

In previous class, we have learnt that the bodies that revolve around the planets are called **satellites**. The moon, therefore, is a satellite of the earth. But since the moon is a natural body, it is a natural satellite of the earth.

Man-made things that revolve around the planets are called **artificial satellites**.

Now-a-days every advanced spacecrafts, provided with a lot of complex machines and technologies, are



An artificial satellite exchanging signals with its headquarters on the earth.

sent into space to revolve around different planets including the Earth, Mars, Mercury, etc.

United States of America has sent a large number of artificial satellites into space by now. India sent its first artificial satellite, **Aryabhata** into space in 1975. Since then, a large number of satellites have been sent into space and are working satisfactorily.



Know More

Russia launched the first artificial satellite Sputnik-1 on October 4, 1957.

An artificial satellite is carried in the space by a long space vehicle called **rocket**. **Polar Satellite Launch Vehicle (PSLV)** and **Geo-Stationary Satellite Launch Vehicle (GSLV)** are India's two main rockets. Satellites can take images of different regions of the planet they revolve around and send it to their centres located on the earth. They can also receive electronic signals and amplify it a large number of times. They have helped a lot in improving the signal availability for television, radio, mobile phones, etc.

Satellites, therefore, have helped the **communication revolution** take place on the earth.



KEY WORDS

Transparent Substance : A substance through which you can see a thing cleanly, is called transparent substance.

Opaque Substance : A substance through which light rays cannot pass, is called an opaque substance.

Translucent Substance : A substance through which light rays can pass partially is called translucent substance.

Artificial Satellites : Man-made things that revolve around the planets are called artificial satellites.



OVERVIEW ON THE CHAPTER

- ▶ Gravitational force of moon causes tides in seas and oceans on the earth.
- ▶ Shadows are formed when light rays are obstructed by some objects.
- ▶ Eclipses occur when the sun, the moon and the earth are in a straight line.
- ▶ A solar eclipse occurs when the moon comes in-between the sun and the earth.
- ▶ A lunar eclipse occurs when the earth comes in-between the sun and the moon.



Assignment



COGNITIVE SKILLS



Multiple choice questions :

(Learning by Doing)

Tick the (✓) correct option.

1. A substance which lets the light rays through itself only partially is called _____.
- a. transparent ☐ b. opaque ☐ c. translucent ☐ d. trans opaque ☐

2. Shadows are shortest _____.
 a. in morning b. at noon c. in evening d. none of these
3. A solar eclipse occurs on a _____ moon day.
 a. new b. half c. full d. partial
4. A lunar eclipse occurs on a _____ moon day.
 a. new b. half c. full d. partial



Fill in the blanks :

Aryabhata opaque low satellite small

1. There lies a _____ tide between two high tides.
2. Shadow of a small object is _____.
3. A body revolving around a planet is called _____.
4. _____ was India's first artificial satellite.
5. We can not see through a _____ object.



Write T for true and F for false :

1. Tides happen due to gravitational force of the sun.
2. Shadows formed in a bright light are bright.
3. Moon is the natural satellite of the earth.
4. Russia's Sputnik - 1 was the first artificial satellite.
5. PSLV and GSLV are two main rockets of India.

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Match the column A with the column B :

A

1. A substance through which things can be seen clearly
2. A substance that lets the light rays pass through it partially
3. A substance through which light rays cannot pass

B

- (a) translucent
- (b) transparent
- (c) opaque



Answer the following questions :

1. What cause tides in seas and oceans ?

2. Why are shadows formed only during day and not during night ?

3. What are the factors that affect the size of the shadows ?

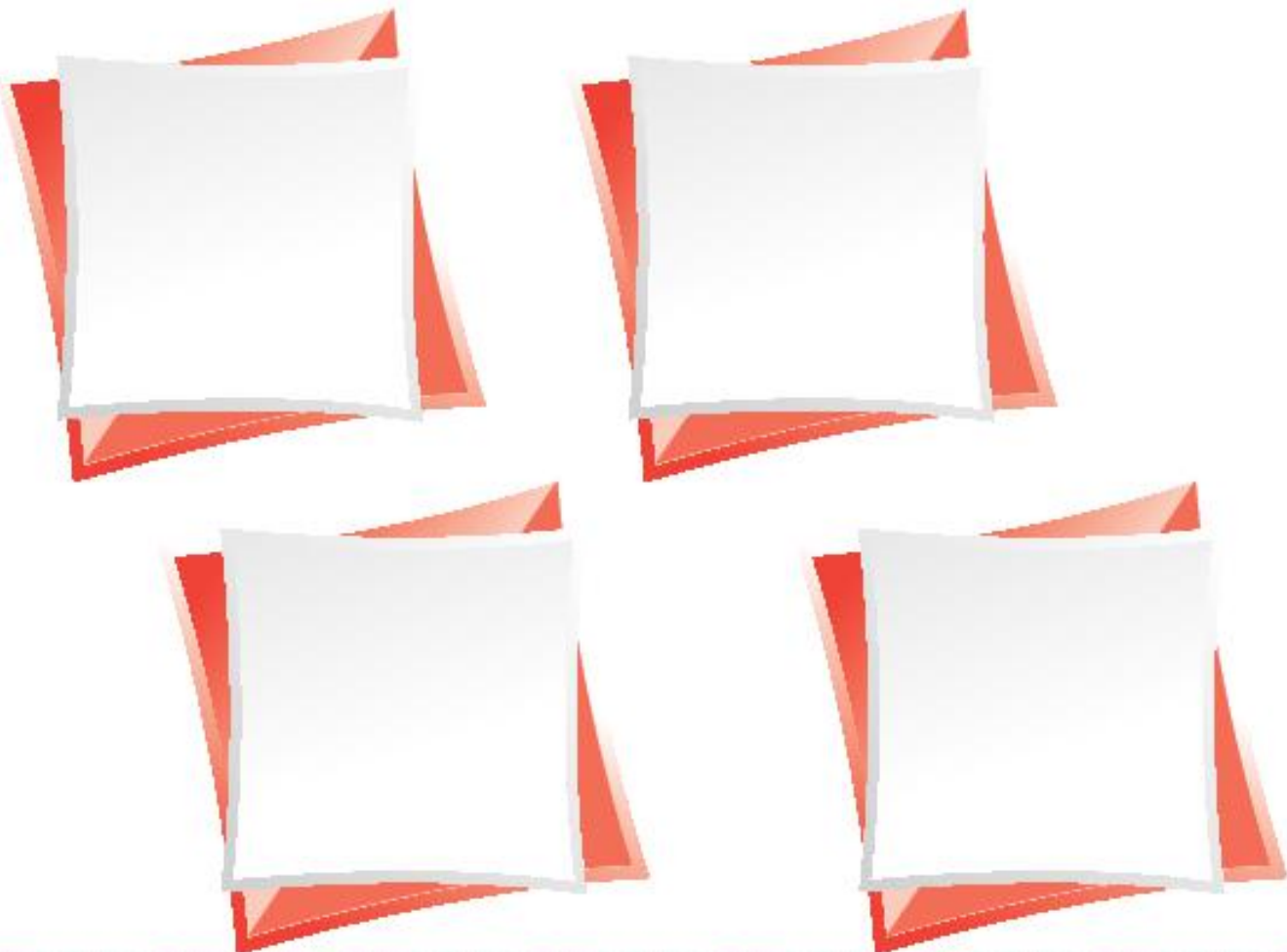
4. When does a solar eclipse take place ?

5. What is an artificial satellite?

ACTIVITY

(Creativity)

Collect the photographs of different artificial satellites of India and paste them in space given.



CRITICAL THINKING

(Self-Awareness)

Write a short note on different kinds of shadow :

**Multiple choice questions :****Tick (✓) the correct option.**

- _____ is the head of a colony.
a. queen bee ☐ b. worker bee ☐ c. drone ☐ d. soldier ant ☐
- _____ eating animals are the primary consumers.
a. plant ☐ b. flesh ☐ c. bird ☐ d. insect ☐
- A matter that cannot flow is _____.
a. gas ☐ b. liquid ☐ c. solid ☐ d. none of these ☐
- A seed is covered by a _____.
a. seed coat ☐ b. embryo ☐ c. cotyledon ☐ d. root ☐
- A place where food grains are stored is called a _____.
a. granary ☐ b. store room ☐ c. cold storage ☐ d. mine ☐

**Fill in the blanks :**

- Nest of a honey bee _____.
- _____ and _____ are dependent on each other.
- Plants and animals breathe _____ dissolved in water.
- Scattering of seeds to far off places is called seed _____.
- _____ and _____ spoil food.

**Answer the following questions :**

- Why are ants and bees called social insects?
- Why are plants called primary producers?
- Why do solids have a definite shape?
- What is inside a seed?
- Name five seeds that we eat.

**Multiple choice questions :**

Tick (✓) the correct option.

- Air enters into the lungs through _____.
a. mouth ☐ b. windpipe ☐ c. nostrils ☐ d. muscles ☐
- The yellow substance inside a bird's egg is called _____.
a. albumen ☐ b. embryo ☐ c. yolk ☐ d. moulting ☐
- The centre part of each tooth _____.
a. dentine ☐ b. pulp ☐ c. crown ☐ d. root ☐
- The only planet where life exists is _____.
a. venus ☐ b. mars ☐ c. earth ☐ d. mercury ☐
- _____ present on the earth's surface is not fit for drinking.
a. water ☐ b. land ☐ c. air ☐ d. none of these ☐

**Fill in the blanks :**

- _____ are the building blocks of life.
- _____ looks like a fish but it is mammal.
- _____ hold our teeth.
- Man uses space probes to study _____.
- Water pollution reduces the _____ present in the water.

**Answer the following questions :**

- Define an organ system.
- Why do birds sit on their eggs?
- Why are teeth important?
- What is universe?
- Write the steps to purify water.